MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

A. Wavelength of microwaves is greater than that of ultraviolet rays.

B. The wavelength of infrared rays is lesser than that of ultraviolet rays.

C. The wavelength of microwaves is lesser than that of infrared rays.

D. Gamma-ray has the shortest wavelength in the electromagnetic spectrum.

What should be the correct statement from the given statements ?

(A) A and B are true

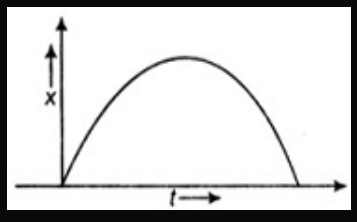
(B) B and C are true

(C) C and D are true

(D) A and D are true

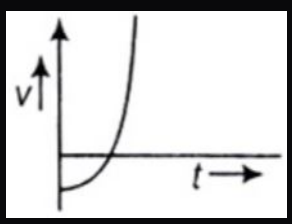
Q.2

The displacement-time $(x - t)$ graph of a body is given as

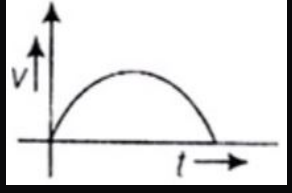


The corresponding velocity-time $(v - t)$ graph will be

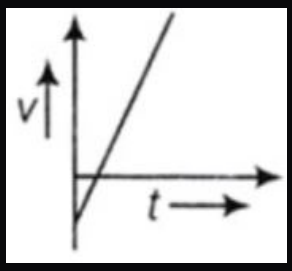
(A)



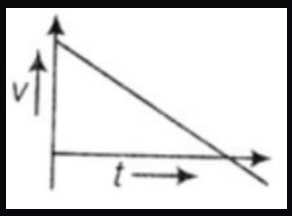
(B)



(C)



(D)



Q.3

In Young's double slit experiment, if the widths of the slit are in the ratio 4 : 9, ratio of intensity of maxima to intensity of minima will be

(A) 25 : 1

(B) 9 : 4

(C) 3 : 2

(D) 81 : 16

Q.4

A charge Q_1 exerts force on a second charge Q_2. If a 3rd charge Q_3 is brought near, the force of Q_1 exerted on Q_2,

(A) Will increase

(B) Will decrease

(C) Will remain unchanged

(D) Will increase if Q_3 is of the same sign as Q_1 and will decrease if Q_3 is of opposite sign

Q.5

The engine of a motorboat is switched off when its speed is v_0. If motorboat experiences variable acceleration as $\frac{dv}{dt} = -\beta v^3$, where β is a positive constant and v instantaneous speed, then the speed of motorboat after t second ($t = 0$ when switch is off) is

(A) $\frac{v_0}{\sqrt{1+2\beta v_0^2 t}}$

(B) $\frac{v_0}{\sqrt{1+\beta v_0 t}}$

(C) $\frac{v_0}{\sqrt{1-\beta v_0^2 t}}$

(D) $\frac{v_0}{\sqrt{1-\beta v_0 t}}$

Q.6

300 cc of a gas is compressed to 150 cc at the atmospheric pressure of 10^6 dyne cm^{-2} . If the change is sudden, what is final pressure? Given $\gamma = 1.4$

(A) 1×10^6 dyne cm^{-2}

(B) 2.0×10^6 dyne cm^{-2}

(C) 2.6×10^6 dyne cm^{-2}

(D) 3.0×10^6 dyne cm^{-2}

Q.7

An object of mass 0.4 kg is whirled in a horizontal circle of radius 2m. If it performs 60 revolutions min^{-1}, calculate the centripetal force acting on it.

(A) 31.58 N

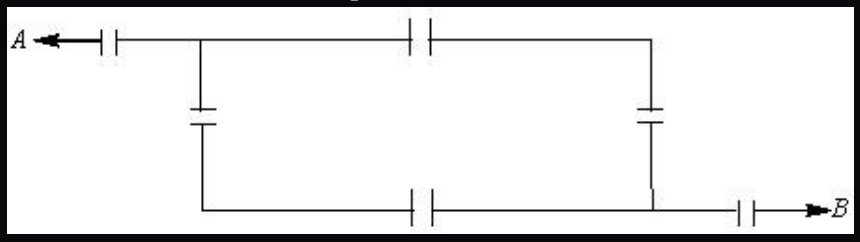
(B) 25.50 N

(C) 40.25 N

(D) 13.58 N

Q.8

A network of six identical capacitors, each of value C, is made, as shown in the figure.



The equivalent capacitance between the points A and B is

(A) $\frac{3C}{4}$

(B) $\frac{3C}{2}$

(C) $3C$

(D) $\frac{C}{3}$

Q.9

An alternating e.m.f. of frequency, $f = \frac{1}{2\pi\sqrt{LC}}$ is applied to a series LCR circuit. For this frequency of the applied e.m.f.

(A) the circuit is at resonance and its impedance don't involve a reactive part.

(B) the current in the circuit is in phase with the applied e.m.f. and the voltage across R equals this applied e.m.f.

(C) the sum of the potential differences across the inductance and capacitance equals the applied e.m.f. which is 180° ahead of phase of the current in the circuit.

(D) the quality factor of the circuit is ω/LR and is a measure of the voltage magnification produced by the circuit at resonance.

Q.10

The ratio of the kinetic energy at mean position to the potential energy at half of the amplitude is

(A) $1:4$

(B) $1:2$

(C) $2:1$

(D) $4:1$

Q.11

The time dependence of a physical quantity p is given by $\mathrm{p} = \mathrm{p}_0 \exp^{(-\alpha t^2)}$ where α is a constant and t is the time. The constant α

(A) has dimensions of p

(B) has dimensions of T^2

(C) has dimensions of T^{-2}

(D) is dimensionless

Q.12

Assertion: Fraunhofer diffraction occurs when all the rays passing through a narrow slit are approximately parallel to one another.
Reason: Fraunhofer diffraction pattern can be achieved by placing the screen far from the slit.

(A) If both assertion and reason are true, but reason is not correct explanation of assertion.

(B) If assertion is true, but reason is false.

(C) If assertion is false, but reason is true.

(D) If both assertion and reason are false.

Q.13

These question consists of two statements each printed as assertion and reason. While answering these question you are required to choose any one of the following five responses.
Assertion: Small water drops are spherical while bigger water drops are not.
Reason: In small water drops surface tension forces dominate while in bigger water drops gravity forces dominate.

(A) If both assertion and reason are correct and reason is the correct explanation of assertion.

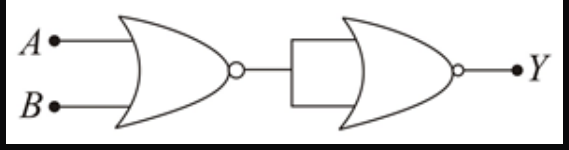
(B) If both assertion and reason are true but reason is not the correct explanation of assertion.

(C) If assertion is true but reason is false.

(D) If assertion is false but reason is true.

Q.14

In the following circuit, the output for Yall possible inputs A and B is expressed by the truth
table :-



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(B)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(C)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(D)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Q.15

If the mean free path of gaseous molecules is 60 cm at a pressure of 1×10^{-4} mm Hg, what will be its mean free-path when the pressure is increased by 100 times ?

(A) 6.0×10^{-1} cm

(B) 6.0 cm

(C) 6.0×10^{1} cm

(D) 6.0×10^{2} cm

Q.16

The ratio of accelerations due to gravity $g_1 \ : \ g_2$ on the surfaces of two planets is $5 \ : 2$ and the ratio of their respective average densities $\rho_1 : \rho_2$ is $2 \ : 1$. What is the ratio of respective escape velocities $v_1 \ : v_2$ from the surface of the planets?

(A) $5 : 2$

(B) $\sqrt{5} : \sqrt{2}$

(C) $5 : 2\sqrt{2}$

(D) $25 : 4$

Q.17

Which of the following statements is wrong?

(A) KE of a body is independent of the direction of motion

(B) In an elastic collision of two bodies ,the momentum and energy of system is conserved

(C) If two protons are brought towards each other the PE of the system increases.

(D) A body cannot have energy without momentum.

Q.18

An electron is moving along a positive $x-$axis. A uniform electric field exists towards negative $y-$axis. What should be the direction of a magnetic field of suitable magnitude, so that the net force of electron is zero.

(A) Positive $z-$axis

(B) Negative $z-$axis

(C) Positive $y-$axis

(D) Negative $y-$axis

Q.19

The binding energy of a H-atom, considering an electron moving around a fixed nucleus (proton), is
$B = -\frac{me^4}{8n^2\varepsilon_0^2h^2} \ \ (m = \text{electron} \ \ \text{mass})$
If one decides to work in a frame of reference where the electron is at rest, the proton would be moving around it. By similar arguments, the binding energy would be
$B = -\frac{Me^4}{8n^2\varepsilon_0^2h^2} \ \ (M = \text{proton} \ \ \text{mass})$
This last expression is not correct, because

(A) n would not be integral.

(B) Bohr-quantisation applies only two-electron.

(C) the frame in which the electron is at rest is not inertial.

(D) the motion of the proton would not be in circular orbits, even approximately.

Q.20

Gas escapes from the surface of a planet because it acquires an escape velocity. The escape velocity will depend on which of the following factors?
I. Mass of the planet.

II. Mass of the particle escaping.
III. Temperature of the planet.
IV. Radius of the planet.
Select the correct answer from the codes given below.

(A) I and II

(B) II and IV

(C) I and IV

(D) I, III and IV

Q.21

The ends of a uniform metal rod of length $100\ \mathrm{cm}$ and area of cross-section $2\ \mathrm{cm}^2$ are maintained at $0\ ^{\circ}$Cand $100\ ^{\circ}\mathrm{C}$. At the mid point of the rod, heat is supplied at a constant rate of $40\ \mathrm{Js}^{-1}$. If the temperature gradient on the higher temperature side of the rod in steady state is $50\ x^{\circ}\ \mathrm{cm}^{-1}$, then the value of x is (Thermal conductivity of the metal $= 400\ \mathrm{J\ s^{-1}\ m^{-1}\ K^{-1}}$)

Q.22

The difference between the radi of nth and $(n+1)^{th}$ orbits of hydrogen atom is equal to the radius of $(\mathrm{n}-1)^{\mathrm{th}}$ orbit of hydrogen. The angular momentum of the electron in the nth orbit is (h is Plank's constant) $\frac{Kh}{\pi}$, find K.

Q.23

A galvanometer has a resistance of $100\ \Omega$. A potential difference of $100\ \mathrm{mV}$ between its terminals gives a full scale deflection. The shunt resistance required to convert it into an ammeter reading upto $5\ \mathrm{A}$, is r ohm, find 200r

Q.24

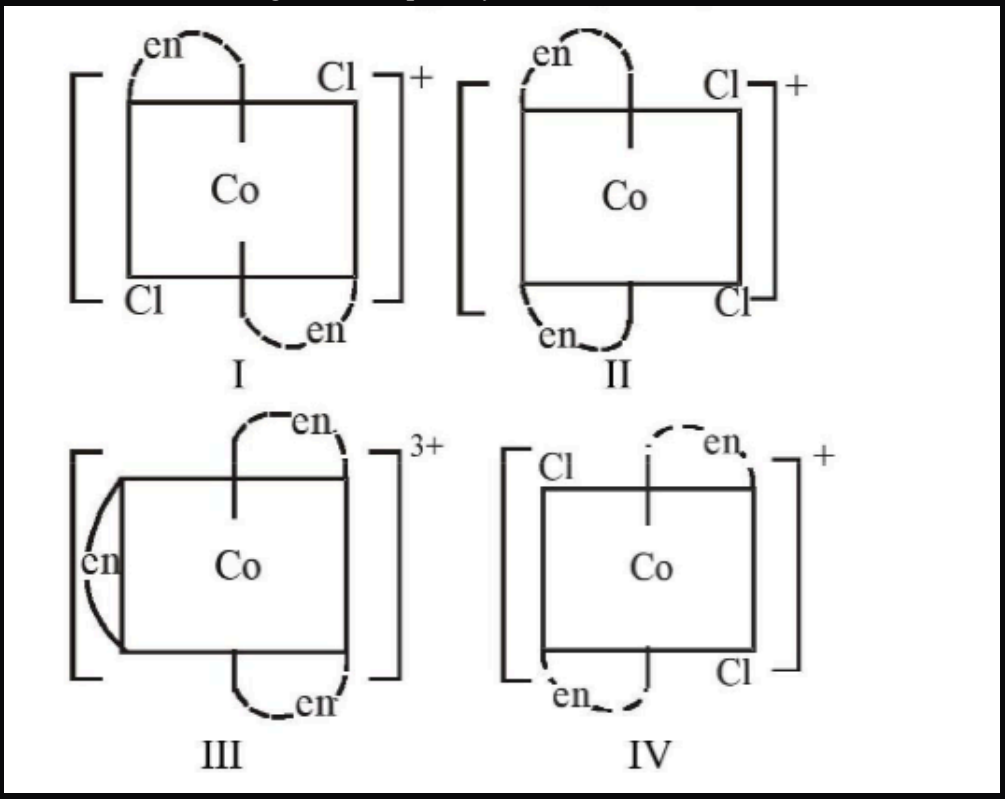
If B is the magnetic field at the centre of a circular loop of area A and its magnetic dipole moment is found to be $n\frac{BA^m}{\mu_0\sqrt{\pi}}$, then what is the value of $10(\mathrm{n}+\mathrm{m})$?

Q.25

An open pipe $40\ \mathrm{cm}$ long, and a closed pipe $31\ \mathrm{cm}$ long, both are having same diameter. Both are in unison in their first overtone. Determine end correction of closed pipe in cm.

Q.26

Which of the following ions are optically active?



(A) I only

(B) II only

(C) II and III

(D) IV only

Q.27

An organic compound (A) reacts with sodium metal and forms (B). On heating with conc. H_2SO_4, (A) gives diethyl ether, (A) and (B) are

(A) C_2H_5OH and C_2H_5ONa

(B) C_3H_7OH and CH_3ONa

(C) CH_3OH and CH_3ONa

(D) C_4H_9OH and C_4H_9ONa

Q.28

Match List I and List II and choose the correct answer by the codes given below the lists.

List I		List II	
A.	Uranium oxide	1.	Red
B.	Cuprous oxide	2.	Blue
C.	Cobalt oxide	3.	Green
D.	Chromium oxide	4.	Yellow

(A)

A	B	C	D
4	1	2	3

(B)

A	B	C	D
3	2	1	4

(C)

A	B	C	D
4	2	1	3

(D)

A	B	C	D
3	1	2	4

Q.29

Which of the following is incorrect?

(A) Molecular weight of NaCl found by osmotic pressure measurements if half of the theoretical value.

(B) Molecular weight of $\mathrm{CH_3\ COOH}$ in benzene found by cryoscopic methods is double of the theoretical value.

(C) Osmotic pressure of $0.1\ \mathrm{M}$ glucose solution is half of that $0.1\ \mathrm{M}\ \mathrm{NaCl}$ solution.

(D) Molecular weight of HCl found by any colligative property will be same in the aqueous solution and benzene solution.

Q.30

Give the reactivity in the decreasing order of the following nucleophiles towards nucleophilic addition reaction with compound $\mathrm{A}\ (\mathrm{F_3C-} \equiv -\mathrm{CF_3})$.

I. $\mathrm{CH_3O^{\ominus}}$

II. $\mathrm{C_2H_5O^{\ominus}}$

III. $\mathrm{CH_3COO^{\ominus}}$

IV. $\mathrm{CH_3SO_3^{\ominus}}$

(1) (II) $>$ (I) $>$ (III) $>$ (IV)

(2) (IV) $>$ (III) $>$ (I) $>$ (II)

(3) (I) $>$ (II) $>$ (IV) $>$ (III)

(4) (III) $>$ (IV) $>$ (II) $>$ (I)

(A) $(\mathrm{II}) > (\mathrm{I}) > (\mathrm{III}) > (\mathrm{IV})$

(B) $(\mathrm{IV}) > (\mathrm{III}) > (\mathrm{I}) > (\mathrm{II})$

(C) $(\mathrm{I}) > (\mathrm{II}) > (\mathrm{IV}) > (\mathrm{III})$

(D) $(\mathrm{III}) > (\mathrm{IV}) > (\mathrm{II}) > (\mathrm{I})$

Q.31

Select the correct option by assigning true (T) and false (F) for the following statements.

(a) In $\mathrm{IF_7}$, iodine is in the super octet state.

(b) $\mathrm{BF_3}$ is a hypervalent compound.

(c) $\pi-$bond is cylindrically symmetrical.

(A) TFF

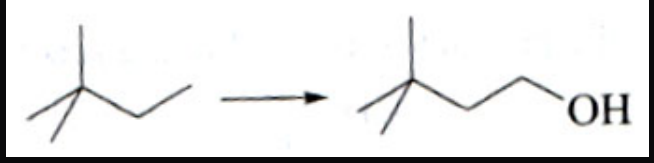
(B) FFF

(C) TTT

(D) FTF

Q.32

What sequence of reagents is required to accomplish the following transformation ?



(A) **(i)** NBS, ROOR ; **(ii)** $CH_3CH_2\overset{\ominus}{O}$ **(iii)** $N\overset{\ominus}{H}_2$;
(iv) Disiamyl borane **(v)** H_2O_2, $O\overset{\ominus}{H}$

(B) **(i)** Cl_2/hv **(ii)** $O\overset{\ominus}{H}$, heat, **(iii)** 2HCl
(iv) $O\overset{\ominus}{H}$, heat ; **(v)** $HgSO_4$, H_2SO_4

(C) (i) NBS, ROOR ; $O\overset{\ominus}{H}$, DMSO

(D) **(i)** Br_2, hv ; **(ii)** t-butoxide ; **(iii)**BH_3. THF, H_2O_2, $O\overset{\ominus}{H}$

Q.33

In H−atom, if 'x' is the radius of the first Bohr orbit, de Broglie wavelength of an electron in the 3rd orbit is:

(A) $3\pi x$

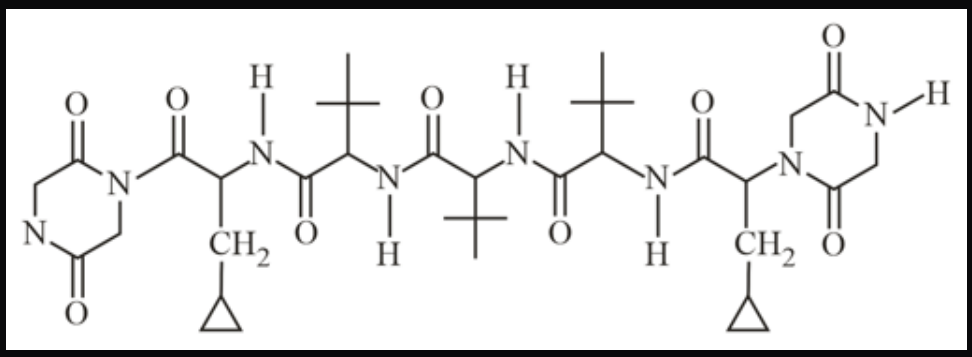
(B) $6\pi x$

(C) $\frac{9x}{2}$

(D) $\frac{x}{2}$

Q.34

The total number of distinct naturally occurring amino acids obtained by complete acidic hydrolysis of the peptide shown below is:



(A) 0

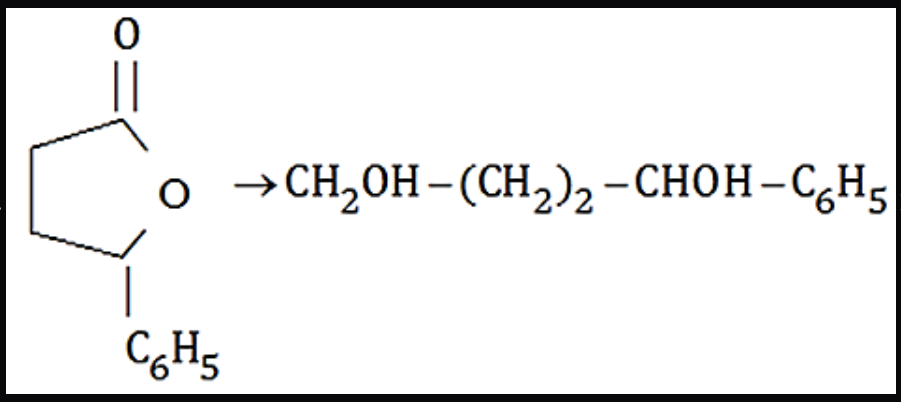
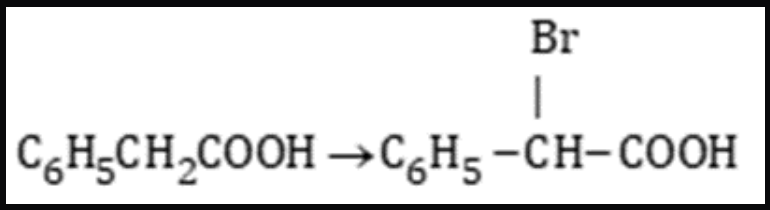
(B) 1

(C) 2

(D) 3

Q.35

Match List I with List II and select the correct answer using the codes given below the lists

List I	List II
P. $CH_3COOC_2H_5 \rightarrow CH_3COCH_2COOC_2H_5$	1. Anhy. $AlCl_3$
Q. 	2. $LiAlH_4$
R. 	3. C_2H_5ONa
S. $C_6H_6 + CH_3COCl \rightarrow C_6H_5COCH_3$	4. P/Br_2

(A) P-3, Q-2, R-1, S-4

(B) P-3, Q-2, R-4, S-1

(C) P-1, Q-2, R-3, S-4

(D) P-4, Q-3, R-2, S-1

Q.36

For the cell reaction
$2Fe^{3+}(aq) + 2I^{-}(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$
$E^0_{cell} = 0.24\,V$ at 298 K. The standard Gibbs energy $(\Delta_r G^{-})$ of the cell reaction is:
[Given that Faraday constant $F = 96500\ Cmol^{-1}$]

(A) $-46.32\ kJ\ mol^{-1}$

(B) $-23.16\ kJ\ mol^{-1}$

(C) $46.32\ kJ\ mol^{-1}$

(D) $23.16\ kJ\ mol^{-1}$

Q.37

Mr. Parker consume 1 kg of bread on average everyday. Bread contain 10% by mass of protein. Protein contain 7% by mass of nitrogen. If 10% of nitrogen present in protein is converted to urea, how much urea is produced in Mr. Parker in one day ?

(A) 750 mg

(B) 70 mg

(C) 1500 mg

(D) 150 mg

Q.38

The end product in the below reaction is

$C_2H_5NH_2 \xrightarrow{HNO_2} A \xrightarrow{PCl_5} B \xrightarrow{NH_3} C$

(A) Ethyl cynide

(B) Ethyl amine

(C) Methyl amine

(D) Acetamide

Q.39

The electronic configurations for some neutral atoms are given below

(I) $1s^2,\ 2s^2\ 2p^6,\ 3s^2$

(II) $1s^2,\ 2s^2\ 2p^6\ ,\ 3s^1$

(III) $1s^2,\ 2s^2\ 2p^6,\ 3s^2\ 3p^2$

(IV) $1s^2,\ 2s^2\ 2p^6,\ 3s^2\ 3p^3$

Which of these is expected to have the highest second ionisation enthalpy?

(A) I

(B) II

(C) III

(D) IV

Q.40

Which of the following organic compounds answers to both iodoform test and Fehling's test?

(A) Ethanol

(B) Methanal

(C) Ethanal

(D) Propanone

Q.41

Two separate experiments were carried out involving adiabatic reversible expansion, taking mono atomic gas in one case and a diatomic gas in the second case. In each case, the initial temperature was T_i and expansion was from volume V_1 to volume V_2 . The final temperatures attained were T_M for mono atomic gas and T_D for diatomic gas. Then

(A) $T_M = T_D < T_i$

(B) $T_M < T_D < T_i$

(C) $T_D > T_M > T_i$

(D) $T_D < T_M < T_i$

Q.42

Statement I : Fluorine cannot be prepared from fluorides by chemical oxidation.

Statement II : Fluorine is the strongest oxidising agent due to its highly positive potential.

(A) Statement I is true, Statement II is also true and Statement II is the correct explanation of Statement I.

(B) Statement I is true, Statement II is also true and Statement II is not the correct explanation of Statement I.

(C) Statement I is true, Statement II is false.

(D) Statement I is false, Statement II is true.

Q.43

Phenol on reaction with a mixture of conc. HNO_3 and conc. H_2SO_4 produces a compound. The degree of unsaturation and the nature of compound is

(A) 5 , acidic

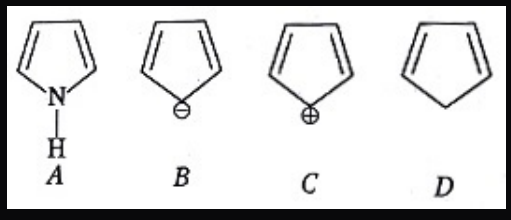
(B) 7 , basic

(C) 6 , neutral

(D) 7 , acidic

Q.44

Which of the following compounds are aromatic?



(A) $A,\ B$

(B) $A,\ B,\ C$

(C) $B,\ C$

(D) $B,\ C,\ D$

Q.45

Which of the following statement(s) is/are not true about the following decomposition reaction?
$2KClO_3 \rightarrow 2KCl + 3O_2$

(i) Potassium is undergoing oxidation
(ii) Chlorine is undergoing oxidation
(iii) Oxygen is reduced
(iv) None of the species are undergoing oxidation or reduction

(A) (i) and (ii)

(B) (i) and (iv)

(C) (i) and (iii)

(D) All of these

Q.46

For the octahedral complexes of Fe^{3+} in SCN^{-}(thiocyanato$-S$) and in CN^{-}ligand environments, the difference between the spin-only magnetic moments in Bohr magnetons (When approximated to the nearest integer) is [Atomic number of $\mathrm{Fe} = 26$]

Q.47

The dissociation constant of a substituted benzoic acid at $25^{\circ}\mathrm{C}$ is 1.0×10^{-4}. The pH of a $0.01\mathrm{M}$ solution of its sodium salt is

Q.48

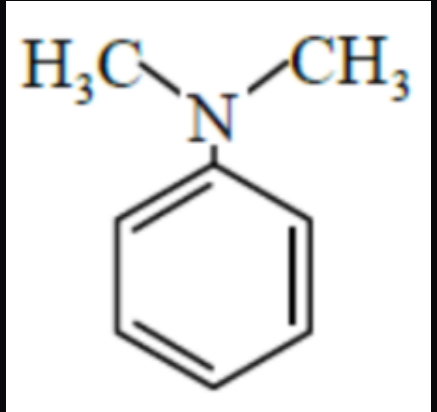
For the 1st order reaction: $\mathrm{A}\,(\mathrm{g}) \rightarrow 2\mathrm{B}\,(\mathrm{g}) + \mathrm{C}\,(\mathrm{s})$; the $\mathrm{t}_{\frac{1}{2}} = 24$ min . The reaction is carried out taking certain mass of A enclosed in a vessel in which it exerts a pressure of 400 mm Hg. The pressure of the reaction mixture in mm Hg after expiry of 48 min will be

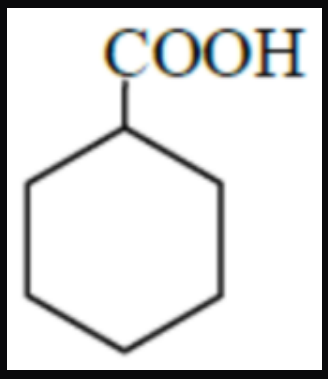
Q.49

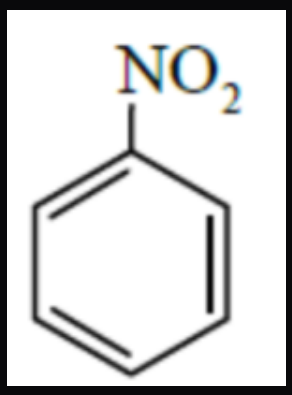
The sum of the number of lone pairs of electrons on each central atom in the following species is $[\mathrm{TeBr}]^{2-}$, $[\mathrm{BrF}_2]^{+}$and $[\mathrm{XeF}_3]^{-}$
(Atomic number: $\mathrm{F} = 9, \mathrm{Br} = 35, \mathrm{Te} = 52, \mathrm{Xe} = 54$)

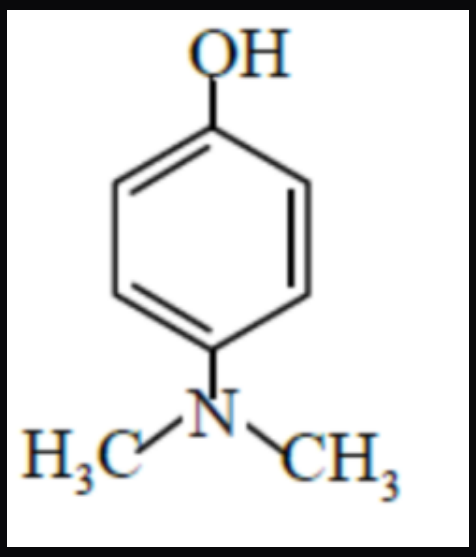
Q.50

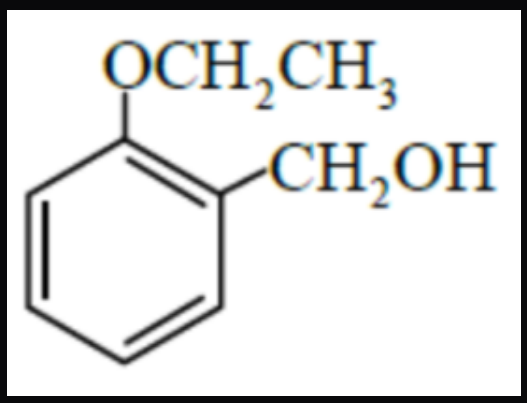
Amongst the following, the total number of compounds soluble in aqueous NaOH is

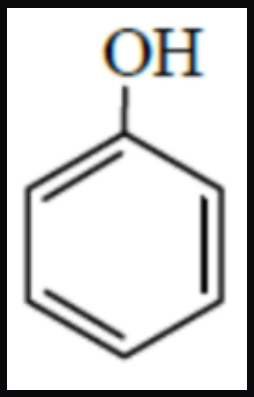


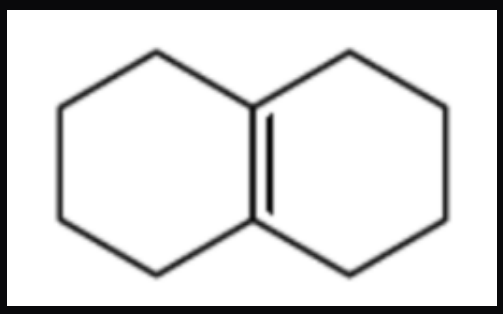

COOH

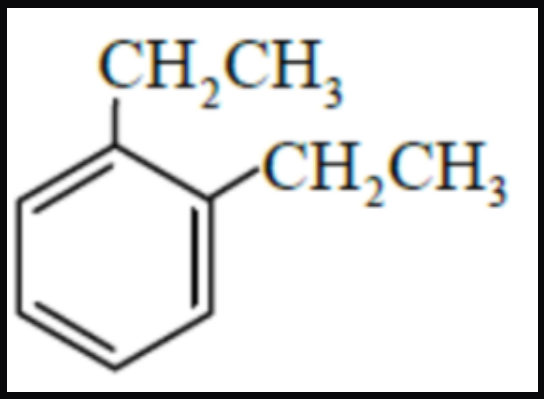

NO2

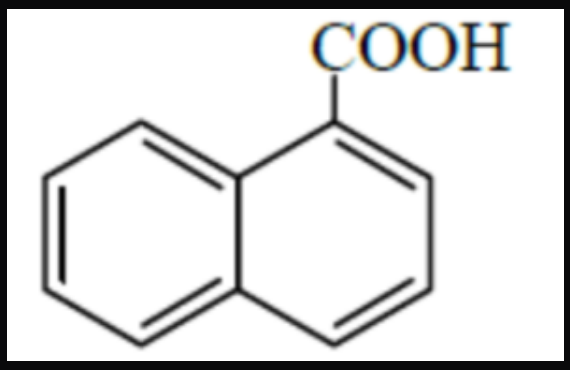

OH
N
H3C
CH3


OCH2CH3
CH2OH









Q.51

If α & β are the roots of the equation $x^2 - p(x+1) - q = 0$, then the value of $\frac{\alpha^2+2\alpha+1}{\alpha^2+2\alpha+q} + \frac{\beta^2+2\beta+1}{\beta^2+2\beta+q}$ is

(A) 2

(B) 1

(C) 0

(D) none of these

Q.52

If the line $\frac{x-2}{-1} = \frac{y+2}{1} = \frac{z+k}{4}$ is one of the angle bisectors of the lines $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and $\frac{x}{-2} = \frac{y}{3} = \frac{z}{1}$, then the value of k is

(A) 1

(B) 2

(C) 4

(D) 8

Q.53

Let S be the set of real values of λ for which the function $f(x) = x^3 - 3(2\lambda - 1)x^2 + 6\lambda x$ has exactly one local maximum and exactly one local minimum, then S can be

(A) $(0,\ 6)$

(B) $(1,\ 4)$

(C) $(-\infty,\ 0)$

(D) $(0,\ \infty)$

Q.54

The number of rational point(s) (a point $(a,\ b)$ is rational, if a and b both are rational numbers) on the circumference of a circle having centre (π,e) is

(A) at most one

(B) at least two

(C) exactly two

(D) infinite

Q.55

If the coefficients of x^2 and x^3 in the expansion of $(3 + kx)^9$ are equal, then the value of k is

(A) $-9/7$

(B) $9/7$

(C) $7/9$

(D) $-7/9$

Q.56

If n satisfies the relation ${}^{n+5}P_{n+1} = \frac{11(n-1)}{2} \times {}^{n+3}P_n$, then number of point of intersection of 'n' straight line is

(A) 18

(B) 28

(C) 21

(D) 10

Q.57

Let $f : \mathrm{R} \to \mathrm{R}$ be a function such that $f\left(x\right) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$, where $[x]$ denotes the greatest integer function. If $I = \int\limits_{-1}^{2} \frac{x.f(x^2)}{f(x+1)+2} dx$, then the value of $2I =$_____

(A) 1

(B) 1.5

(C) 0.5

(D) 2.5

Q.58

If $f(x) = \max\left\{\frac{x}{n},\ |\sin \pi x|\right\},\ n \in N$ has maximum points of non-differentiability for $x \in (0,\ 4)$, then n cannot be

(A) 4

(B) 2

(C) 5

(D) 6

Q.59

Let $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $f(x) = \sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2+x}}, x > 0$. If the least value of $f(x)$ is $2\sqrt{3}$, then α is equal to

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{8}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{4}$

Q.60

f is a function from $A \to A$ where $A = \{1, 2, 3,\ \ldots, 10\}$. If all the discrete points on the graph of $f(x)$ are joined then the probability that f is onto and the resulting graph has only one local minima and no local maxima is $\frac{\lambda}{10^9}$, then the number of factors of λ is :

(A) 4

(B) 2

(C) 3

(D) 1

Q.61

A car is parked by an owner amongst 25 cars in a row not at the two extremes. On his return, he find that exactly 15 places are still occupied. The probability that both the neighbouring places are vacant is

(A) $\frac{15}{92}$

(B) $\frac{91}{276}$

(C) $\frac{15}{184}$

(D) None of these

Q.62

Consider the matrix $A = \begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix}$, then $(\mathrm{I} + A)^{40}$ is equal to

(A) $\mathrm{I} + 2^{38}A$

(B) $\mathrm{I} + 2^{39}A$

(C) $\mathrm{I} + \left(2^{40} + 1\right)A$

(D) $\mathrm{I} + \left(2^{40} - 1\right)A$

Q.63

The area (in sq. units) bounded by $[|x|] + [|y|] = 2$ in the first and third quadrant is (where, $[.]$ is the greatest integer function),

(A) 4

(B) 3

(C) 6

(D) 10

Q.64

$ABCD$ is a quadrilateral. E is the point of intersection of the line joining the mid-points of the opposite sides. If O is any point and $\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} + \overrightarrow{OD} = x\overrightarrow{OE}$, then x is equal to

(A) 3

(B) 9

(C) 7

(D) 4

Q.65

Let $A,\ B$ and C are $n \times n$ matrices such that $|A| = 2,\ |B| = 3$ and $|C| = 5$. If $\left|(2A)^2(3B)(5C)^{-1}\right| = \frac{1728}{125}$, then the value of n is equal to

(A) 2

(B) 3

(C) 4

(D) 6

Q.66

The orthocentre of the triangle with vertices $(-2,\ -6),\ (-2,\ 4)$ and $(1,\ 3)$ is

(A) $(-3,\ 1)$

(B) $(-1,\ 1/3)$

(C) $(1,\ 3)$

(D) None of these

Q.67

Given $i = \sqrt{-1}$, then the value of the sum $\frac{1}{1+i} + \frac{1}{1-i} + \frac{1}{-1+i} + \frac{1}{-1-i} + \frac{2}{1+i} + \frac{2}{1-i} + \frac{2}{-1+i} + \frac{2}{-1-i} + \frac{3}{1+i} + \frac{3}{1-i} + \frac{3}{-1+i} + \frac{3}{-1-i} + \ldots + \frac{n}{1+i} + \frac{n}{1-i} + \frac{n}{-1+i} + \frac{n}{-1-i}$ is

(A) $2n^2 + 2n$

(B) $2in^2 + 2in$

(C) $(1+i)n^2$

(D) None of these

Q.68

$\lim\limits_{x \to \frac{\pi}{4}} \frac{\tan x - 1}{\cos 2x}$ is equal to

(A) 1

(B) 0

(C) -2

(D) -1

Q.69

The direction cosines of two lines are given by $2\ell + \lambda \mathrm{m} + \mathrm{n} = 0$ and $\mathrm{mn} + \mathrm{n}\ell - 5\ell\mathrm{m} = 0$. Find value of λ,if the lines are perpendicular.

(A) $\frac{1}{3}$

(B) $\frac{2}{9}$

(C) $\frac{2}{3}$

(D) None of these

Q.70

$A,\ B,\ C$ are three matrices of the same order such that any two are symmetric and the 3rd one is Skew-Symmetric. If $X = ABC + CBA$ and $Y = ABC - CBA$, then $(XY)^T$ is

(A) Symmetric

(B) Skew-Symmetric

(C) $I - XY$

(D) $-YX$

Q.71

If $\int_0^{\pi} \sqrt{\left(\cos x+\cos\ 2x+\cos\ 3x\right)^2+(\sin x+\sin\ 2x+\sin\ 3x)^2}\ dx$ has the value equal to $\left(\frac{\pi}{k}+\sqrt{w}\right)$ where k and w are positive integers, find the value of $\left(k^2+w^2\right)$.

Q.72

Find the number of real values of m for which $A \cup B$ has exactly 3 distinct elements, where A and B are following two sets:
$A=\{x: x^2+(m-1)x-2(m+1)=0,\ x\ \in R\}$
$B=\{x:(m-1)x^2+mx+1=0,\ x\ \in R\}$

Q.73

If the line joining the foci of the hyperbola $S_1 \equiv \frac{x^2}{a^2}-\frac{y^2}{b^2}+1=0$ does not subtend a right angle at any point on the hyperbola $S_2 \equiv \frac{x^2}{4a^2}-\frac{y^2}{b^2}=1$, then number of integral values of $4e^2$ is/are (e is eccentricity of $S_2=0$)

Q.74

Let $S=\{1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7\}$. Then the number of possible functions $f: S \to S$ such that $f(m \cdot n)=f(m) \cdot f(n)$ for every $m,\ n \in S$ and $m \cdot n \in S$ is equal to _____.

Q.75

The first term of a sequence is 2014 . Each succeeding term is the sum of the cubes of the digits of the previous term. Then the 2014th term of the sequence is

Answers & Solutions

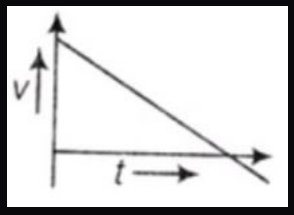
Q.1 Answer:

A and D are true

Solution:

Microwaves have a longer wavelength than UV rays. Since the frequency of microwaves is smaller than that of ultraviolet rays, the wavelength of microwaves would be longer.
Gamma-rays have the highest frequency in the electromagnetic spectrum. As a consequence, it has the shortest wavelength.
Infrared rays have a longer wavelength than ultraviolet rays, and microwave rays have a longer wavelength than ultraviolet rays. Hence, A and D are correct.

Q.2 Answer:



Solution:

As we known that, the velocity of a body is given by the slope of displacement - time graph. So initially slope of the graph is positive and after sometimes, it becomes zero and then will be negative.

Q.3 Answer:

25 : 1

Solution:

As ratio of slit widths = Ratio of intensities

$\therefore \frac{I_1}{I_2} = \frac{9}{4}$ **or** $\frac{a_1^2}{a_2^2} = \frac{9}{4}$ **or** $\frac{a_1}{a_2} = \frac{3}{2}$

$a_{max.} = a_1 + a_2 = 3 + 2 = 5\ ;\ a_{min.} = 3 - 2 = 1$

$\frac{I_{max.}}{I_{min.}} = \frac{(a_1+a_2)^2}{(a_1-a_2)^2} = \frac{(3+2)^2}{(3+2)^2} = \frac{25}{1}$

Q.4 Answer:

Will remain unchanged

Solution:

According to principle of superposition force acting between the two charges does'nt depend on the presence of other.

Q.5 Answer:

$\frac{v_0}{\sqrt{1+2\beta v_0^2 t}}$

Solution:

$\frac{dv}{dt} = -\beta v^3$

$\int_{v_0}^{v} \frac{dv}{v^3} = -\beta \int_0 dt$

$\frac{1}{-2}\left[\frac{1}{v^2} - \frac{1}{v_0^2}\right] = -\beta t$

$\frac{1}{v^2} = \frac{1+2\beta v_0^2 t}{v_0^2}$

$v = \frac{v_0}{\sqrt{1+2\beta v_0^2 t}}$

Q.6 Answer:

2.6×10^6 dyne cm^{-2}

Solution:

As the process is sudden, so it must be an adiabatic process.

And for adiabatic process, equation of state is,

$P_1 {V_1}^{\gamma} = P_2 {V_2}^{\gamma}$

$\Rightarrow P_2 = \frac{P_1 {V_1}^{\gamma}}{{V_2}^{\gamma}}$

Substituting all the given values, we have,

$\Rightarrow P_2 = \left(10^6\right)\left(\frac{300}{150}\right)^{1.4} = 2.6 \times 10^6$ dyne cm^{-2}

Q.7 Answer:

31.58 N

Solution:

Here, $M = 0.4$ kg ; $r = 2$ m ; $v = 60$ r.p.m. $= 1$r.p.s.

Now, $\omega = 2\pi v = 2\pi \times 1 = 2\pi$ rad s^{-1}

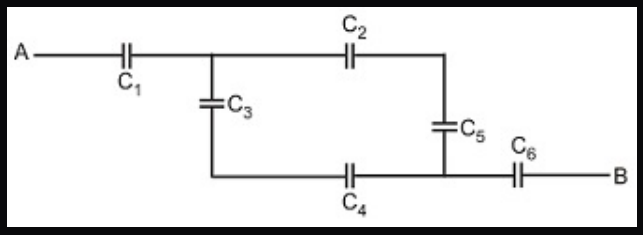
The centripetal force is given by

$\mathrm{F} = \mathrm{M}\ r\ \omega^2 = 0.4 \times 2 \times (2\pi)^2 = 31.58\mathrm{N}$

Q.8 Answer:

$\frac{C}{3}$

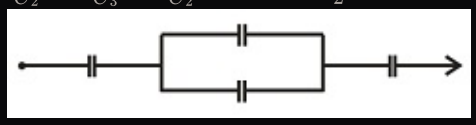
Solution:

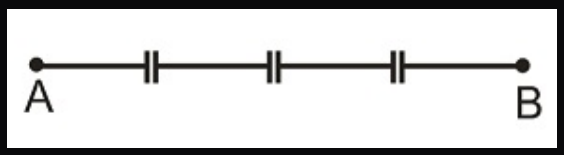


$\frac{1}{C_1\prime} = \frac{1}{C_3} + \frac{1}{C_4} = \frac{2}{C}$

$C_1\prime = \frac{C}{2}$

$\frac{1}{C_2} = \frac{1}{C_3} + \frac{1}{C_2} \Rightarrow C_2 = \frac{C}{2}$





$C_2\text{'} \parallel C_1\text{'} = \frac{C}{2} + \frac{C}{2} = C$

$\frac{1}{C_3\prime} = \frac{1}{C} + \frac{1}{C} = \frac{3}{C} \Rightarrow \text{equation} = \frac{C}{3}.$

Q.9 Answer:

the current in the circuit is in phase with the applied e.m.f. and the voltage across R equals this applied e.m.f.

Solution:

Power factor of an LCR **series circuit is given by,**

$\cos(\phi) = \frac{R}{Z} = \frac{R}{\sqrt{\left(R^2 + (X_\mathrm{c} - X_\mathrm{L})^2\right)}},$

where $R, Z,\ X_\mathrm{c}$ and X_L **represents resistance, impedance, capacitive reactance and inductive reactance, respectively.**

At resonance $X_\mathrm{L} = X_\mathrm{c}$**, on putting it in above relation, we have,**

$\Rightarrow \cos(\phi) = \frac{R}{\sqrt{\left(R^2 + (X_\mathrm{L} - X_\mathrm{L})^2\right)}}$

where f, C and L **represents frequency, capacitance and inductance, respectively.**

$\Rightarrow \cos(\phi) = \frac{R}{\sqrt{\left(R^2 + (0)^2\right)}}$

$\Rightarrow \cos(\phi) = 1$

$\Rightarrow \phi = 0^\circ$

Above value shows in the given circuit, phase angle between voltage and current is 0°**, so the voltage and current are in phase for resonant** LCR **circuit, and thus voltage across** R **will the applied emf.**

Q.10 Answer:

4 : 1

Solution:

Suppose ω is angular velocity and A is amplitude of SHM.

Velocity at a distance x from mean position in SHM is given by-

$v = \omega\sqrt{A^2 - x^2}$

$\therefore$ **Kinetic energy** $(KE) = \frac{1}{2}\mathrm{mv}^2 = \frac{1}{2}m\omega^2\left(A^2 - x^2\right)$

At mean position, $x = 0$

$\therefore\ KE = \frac{1}{2}m\omega^2A^2$

Potential energy is given by,

$PE = \frac{1}{2}m\omega^2x^2$

At half the amplitude, $x = \frac{A}{2}$

$\Rightarrow PE = \frac{1}{4} \times \frac{1}{2}m\omega^2A^2$

$\therefore\ \frac{KE}{PE} = \frac{A^2 - x^2}{x^2}$

Ratio of kinetic energy at mean position and potential energy at half of the amplitude,

$\therefore\ \frac{KE}{PE} = 4$

Q.11 Answer:

has dimensions of T^{-2}

Solution:

Applying the principle of homogeneity of dimensions, we get –

$e^{-\alpha t^2}$ **is dimensionless**

$\therefore\ \alpha t^2$ **is dimensionless**

$\therefore\ \alpha$ **should have dimensions of inverse of time-squared –**

or $[\alpha] = [T^{-2}]$

Q.12 Answer:

If both assertion and reason are true, but reason is not correct explanation of assertion.

Solution:

Fraunhofer diffraction occurs when all the rays are parallel. It does not depend on the spacing between screen and the slit. To achieve parallel rays we can place screen far away from source or slits

Q.13 Answer:

If both assertion and reason are correct and reason is the correct explanation of assertion.

Solution:

Liquid has a tendency to acquire minimum surface area due to surface tension. In gravity free space a drop of any size will be spherical.

In case of gravity, bigger drop will have larger force of gravity which tends to distort the spherical shape of the drop.

Hence, both assertion and reason are correct and reason is the correct explanation of assertion

Q.14 Answer:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Solution:

For first NOR gate, boolean expression is written as

$Y\prime = \left(\overline{A+B}\right)$.

For second NOR gate, boolean expression is written as

$Y = \left(\overline{Y\prime + Y\prime}\right) Y = \overline{Y\prime} \cdot \overline{Y\prime} = \overline{\left(\overline{A+B}\right)} \cdot \overline{\left(\overline{A+B}\right)} = (A+B) \cdot (A+B)$

Truth table of the output will look like as given

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Q.15 Answer:

6.0×10^{-1} cm

Solution:

Relation of mean free path with temperature and pressure:

$\lambda \propto \frac{\mathrm{T}}{\mathrm{P}}$

At constant temperature the relation of mean free path and pressure is:

$\frac{\lambda_2}{\lambda_1} = \frac{\mathrm{P}_1}{\mathrm{P}_2}$

$\lambda_2 = 60 \times \frac{1}{100} \quad = 6 \times 10^{-1}$ cm

Q.16 Answer:

$5 : 2\sqrt{2}$

Solution:

$\because$ **Escape velocity from a planet,**

$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2GM}{R^2} R}$

$= \sqrt{2gR}$ **..... (i)**

Acceleration due to gravity,

$g = \frac{GM}{R^2} = \frac{G\frac{4}{3}\pi R^3 \cdot \rho}{R^2} = \frac{4}{3} G\pi R\rho$

$\therefore \quad$ Radius $R = \frac{3g}{4\pi G\rho}$ **..... (ii)**

From Equations. (i) and (ii), we get

$v_e = \sqrt{2g \cdot \frac{3g}{4\pi G\rho}} = \sqrt{\frac{3}{2} \frac{g^2}{\pi G\rho}}$

Thus, $v_e \propto \frac{g}{\sqrt{\rho}}$

$\therefore \quad \frac{v_{e_1}}{v_{e_2}} = \frac{g_1}{\sqrt{\rho_1}} \times \frac{\sqrt{\rho_2}}{g_2} = \frac{5}{2} \times \frac{1}{\sqrt{2}}$

$\left(\text{given, } \frac{g_1}{g_2} = \frac{5}{2} \text{ and } \frac{\rho_1}{\rho_2} = 2:1\right)$

$= \frac{5}{2\sqrt{2}}$

Q.17 Answer:

A body cannot have energy without momentum.

Solution:

If momentum is Zero ie, if p=0,then kinetic energy

$K = \frac{p^2}{2m} = 0$

But potential energy cannot be zero, thus a body can have energy without momentum.

Q.18 Answer:

Negative $z-$axis

Solution:

$\overrightarrow{F_e} + \overrightarrow{F_m} = 0 \Rightarrow \quad q\overrightarrow{E} + q\left(\overrightarrow{V} \times \overrightarrow{B}\right) = 0 \Rightarrow \quad \overrightarrow{E} + \overrightarrow{V} \times \overrightarrow{B} = 0$

$E\left(-\hat{\mathrm{j}}\right) + VB\left(\hat{\mathrm{i}} \times \widehat{n}\right) = 0 \Rightarrow \quad \hat{\mathrm{i}} \times \widehat{n}$ **should be in** $\hat{\mathrm{j}}$**.** $\Rightarrow \widehat{n} = -\widehat{\mathrm{k}}$

Q.19 Answer:

the frame in which the electron is at rest is not inertial.

Solution:

An electron rotating around a fixed proton in a hydrogen atom exhibits centripetal acceleration. As a result, it has no inertial frame of reference.

The mass of an electron is negligible if compared to that of a proton. Thus, the electrostatic force cannot be produced by the centripetal force.

When one decides to work in a frame of reference where the electron is at rest, the given expression $B = -\frac{Me^4}{8n^2\varepsilon_0^2h^2} \quad (M = \text{proton} \quad \text{mass})$ **is not true as it forms the non-inertial frame of reference.**

Q.20 Answer:

I and IV

Solution:

The formula of escape velocity is,

$v_{\mathrm{e}} = \sqrt{\frac{2\,\mathrm{GM}}{R}}$

Hence, from above equation it is clear that the escape velocity depends upon the mass and radius of the planet.

Q.21 Answer:

—

Solution:

Let the rate of heat flow from point C **to** A **and** C **to** B **is** H_1 **and** H_2 **respectively.**

image

In steady state,

$H_1 + H_2 = 40$

$\Rightarrow \ \frac{\Delta T}{R_1} + \frac{\Delta T}{R_2} = 40$

$\Rightarrow \ \frac{T-0}{\frac{0.5}{400\times2\times10^{-4}}} + \frac{T-100}{\frac{0.5}{400\times2\times10^{-4}}} = 40$

$\Rightarrow \ 2\,\mathrm{T} - 100 = 40 \times \frac{0.5}{400\times2\times10^{-4}}$

$\Rightarrow \ T = \ 175\ ^{\circ}\mathrm{C}$

Temperature gradient on higher temperature side $= \frac{175-100}{\frac{1}{2}} = 150^{\circ}\ \mathrm{cm}^{-1}$

Q.22 Answer:

—

Solution:

The radius of the hydrogen atom,

$r_n = r_0 n^2$

$\Rightarrow r_{n+1} = r_0(n+1)^2,\ r_n = r_0(n)^2\ \&\ r_{n-1} = r_0(n-1)^2 \Rightarrow r_{n+1} - r_n = r_{n-1} \Rightarrow r_0(n+1)^2 - r_0(n)^2 = r_0(n-1)^2 \Rightarrow (n$

now Recall the Bohr's second postulate,

$mvr = \overrightarrow{J} = \frac{nh}{2\pi} = \frac{4h}{2\pi} = \frac{2h}{\pi}$**, where** $\overrightarrow{J}$ **is known as total angular momentum,**

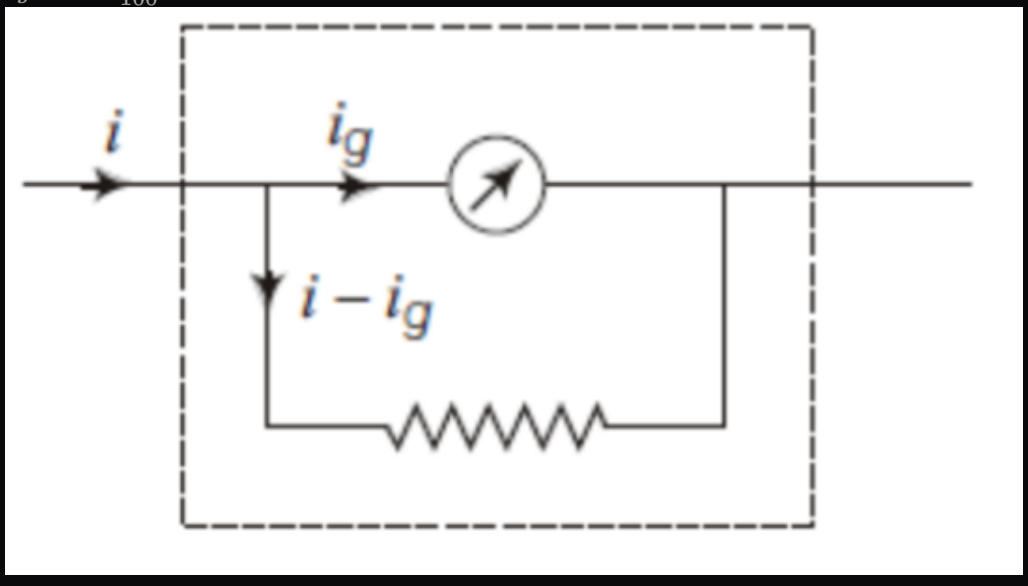
Q.23 Answer:

—

Solution:

Current required in full scale deflection is potential difference across $i_g = \frac{\text{galvanometer}}{\text{resistance of galvanometer}}$

$i_g = \frac{100\times10^{-3}}{100} = 10^{-3}\ \mathrm{A}$



The required shunt resistance is

$S = \left(\frac{i_g}{i-i_g}\right) G$

$= \frac{10^{-3}}{5-10^{-3}} \times 100$

$= 0.02\Omega$

Q.24 Answer:

—

Solution:

Let r be the radius of the circular loop.

$\therefore \quad A = \pi r^2$

$\Rightarrow \mathrm{r} = \sqrt{\frac{\mathrm{A}}{\pi}}$

Magnetic field at the centre of the loop is

$B = \frac{\mu_0 I}{2r}$

$\Rightarrow \mathrm{B}$

$= \frac{\mu_0 \mathrm{I}}{2\sqrt{\frac{\mathrm{A}}{\pi}}} \Rightarrow \mathrm{I} = \frac{2\mathrm{B}}{\mu_0}\sqrt{\frac{\mathrm{A}}{\pi}}$

Now, magnetic moment of the loop is

$\mathrm{M} = \mathrm{IA}$

$= \frac{2\mathrm{B}}{\mu_0}\sqrt{\frac{\mathrm{A}}{\pi}} \cdot \mathrm{A}$

$= \frac{2\mathrm{B}}{\mu_0}\frac{\mathrm{A}^{\frac{3}{2}}}{\sqrt{\pi}}$

$\Rightarrow n = 2$ **and** $m = 1.5$

$\therefore \; n + m = 3.5$

Q.25 Answer:

—

Solution:

$f = \frac{v}{40+2e} = \frac{3v}{4(31+e)}$

$\Rightarrow \quad \frac{3}{4}\left(40 + 2e\right) = 31 + e$

$\Rightarrow 120 + 6e = 124 + 4e$

$2e = 4$

$e = 2 \ \mathrm{cm}$

Q.26 Answer:

II and III

Solution:

Ions I and IV are the same (trans), with mirror plane through en groups.

Q.27 Answer:

$\mathrm{C_2H_5OH}$ and $\mathrm{C_2H_5ONa}$

Solution:

$$\underset{\text{Ethanol}}{\underset{(\mathrm{A})}{2\mathrm{C_2H_5OH}}} + 2\mathrm{Na} \longrightarrow \underset{\text{Sod. ethoxide}}{\underset{(\mathrm{B})}{2\mathrm{C_2H_5ONa}}} + \mathrm{H_2}\uparrow$$

$$2\mathrm{C_2H_5OH} \xrightarrow[\text{conc. } \mathrm{H_2SO_4}]{-\mathrm{H_2O}} \underset{\text{Diethyl ether}}{\mathrm{C_2H_5O \cdot C_2H_5}}$$

Thus, (A) is ethanol and (B) is sodium ethoxide.

Q.28 Answer:

A	B	C	D
4	1	2	3

Solution:

Cuprous oxide (Cu_2O) is one of the major oxides of copper and is red in colour. It is the reduced form of cupric oxide which is black in colour.
Cobalt oxide (CoO) is bluish-green in colour and is used as a colouring agent of glass.
Chromium oxide (Cr_2O_3) is green in colour and is also known as chrome green. It is present in the form of fine crystals.
Uranium oxide is found in several colours ranging from yellow to green. Uranium dioxide also known as urania is yellow in colour.

Q.29 Answer:

Molecular weight of HCl found by any colligative property will be same in the aqueous solution and benzene solution.

Solution:

In aqueous solution, HCl dissociates so molecular weight decreases but in benzene it does not dissociates so molecular weight remains same.

Q.30 Answer:

(II) > (I) > (III) > (IV)

Solution:

In the same nucleophilic centre, nucleophlilicity is parallel to basic character. Stronger the acid, weaker is the conjugated base.
Acidic character order: $CH_3SO_3H > CH_3COOH > CH_3OH > C_2H_5OH$
Basic and nucelophilicity order:

$$\underset{(IV)}{CH_3\overset{\ominus}{S}O_3} < \underset{(III)}{CH_3CO\overset{\ominus}{O}} < \underset{(I)}{CH_3\overset{\ominus}{O}} < \underset{(II)}{C_2H_5\overset{\ominus}{O}}$$

Q.31 Answer:

TFF

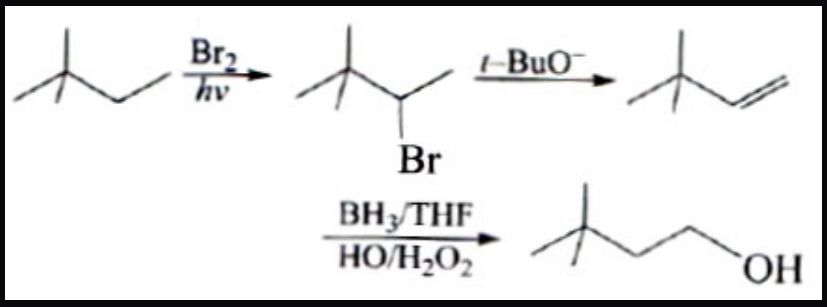
Solution:

The central atom in a super octet molecule contains more than an octet of electrons. Examples are IF_7, ClF_3 and SF_6.
(a) The central atom I in IF_7 has 14 electrons, so, it is in the super octet state.
Hypervalent compounds are the chemical compounds containing a central atom having more than eight electrons in the valence electron shell.
Hypovalent compounds are the chemical compounds containing a central atom with less than eight electrons in the valence electron shell.
(b) The central atom B in BF_3 has 6 electrons, hence, it is a hypovalent compound.
A $\pi-$bond is generated by the overlapping of two $p-$orbitals that are not radially symmetric.
(c) $\pi-$bond is an unsymmetrical bond.

Q.32 Answer:

(i) Br_2, hv ; **(ii)** t-butoxide ; **(iii)**BH_3. THF, $H_2O_2, \overset{\ominus}{O}H$

Solution:



Q.33 Answer:

6 π x

Solution:

According to the Bohr model, the first radius for hydrogen atom:

$r_1 = 0.529A^\circ = x$

The third radius of the Bohr model: $r_3 = 0.529\ n^2\ A^\circ\ =\ 0.529(3)^2A^\circ = 9x$

Hence, the de Broglie wavelength:

$\lambda = \frac{2\pi r_3}{n} = \frac{2\times\pi\times(9x)}{3} = 6\pi x$

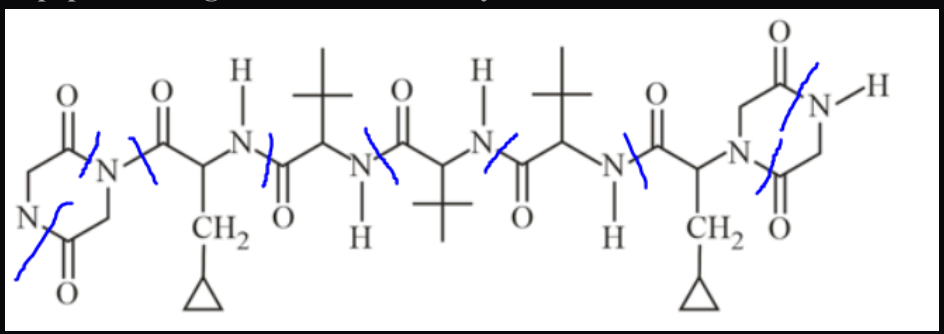
Q.34 Answer:

1

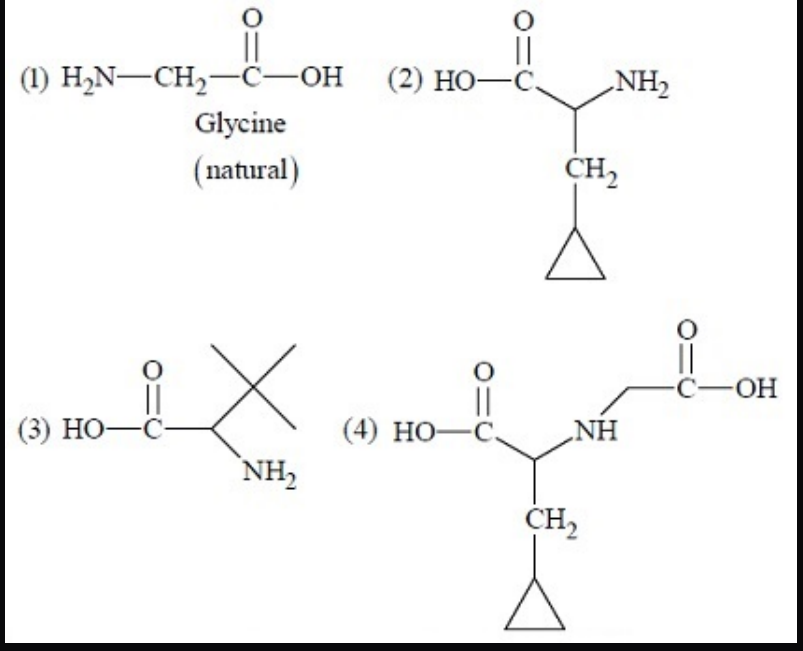
Solution:

Amino acids are the organic molecules which consist of a base i.e., amine group, an acid i.e., carboxylic group and an alkyl group which is different in every amino acid. When the amino acids combine they form a long chain which is known as protein.

So when the given compound will undergo the acidic hydrolysis it gives us four types of amino acids by breakage of peptide linkages, as shown below by blue lines.



We will get four amino acids as shown below.



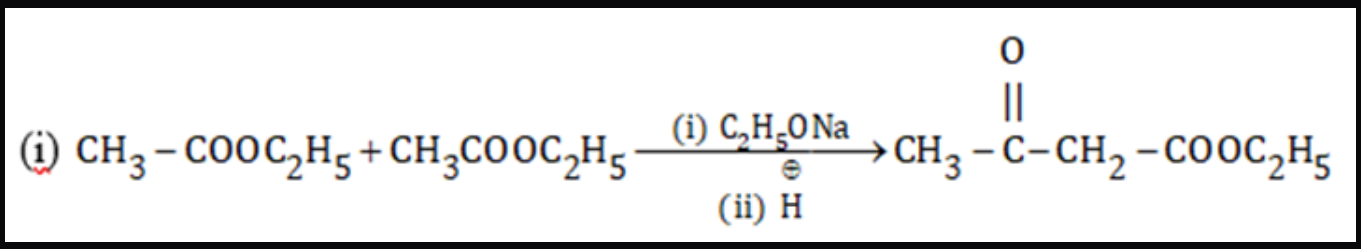
Here the first compound formed is known as glycine where along with the amine group and carboxylic acid group, hydrogen is present as a side chain. It is the only amino acid formed which is found naturally in the environment. Whereas, the other amino acids formed are synthesised artificially.

Hence, there will be only one natural amino acid.

Q.35 Answer:

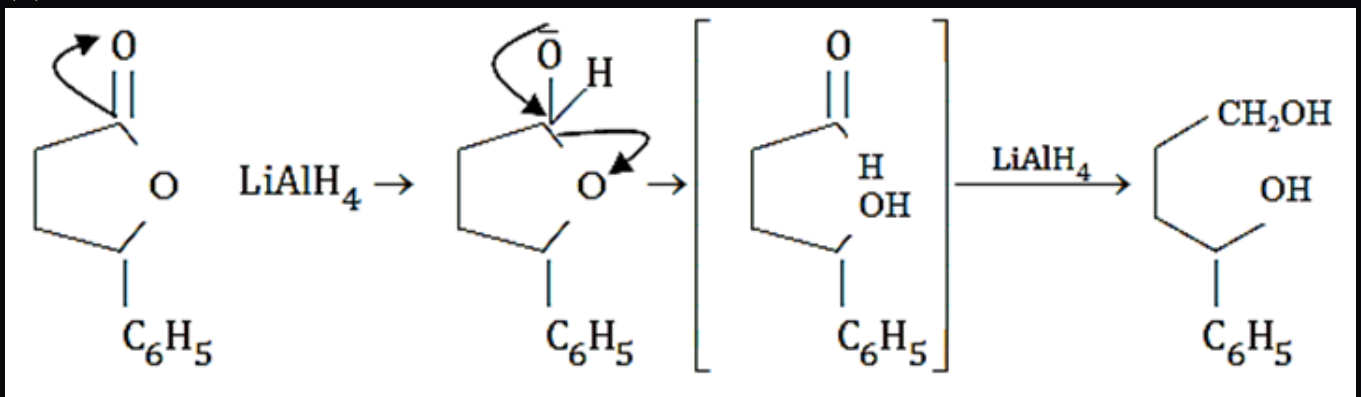
P-3, Q-2, R-4, S-1

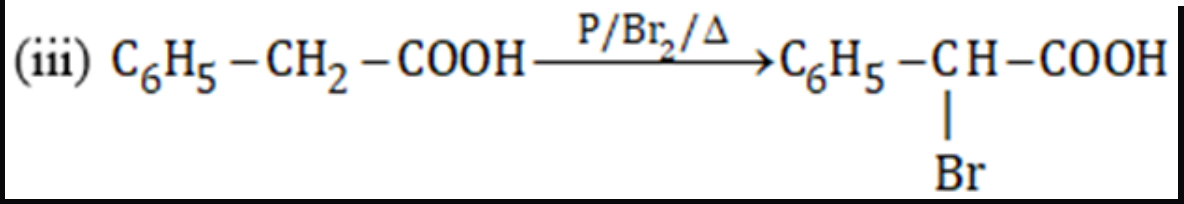
Solution:



The reaction is Claisen condensation.

(ii)





(iv) $C_6H_6 + CH_3COCl \xrightarrow{\text{Anhy.}AlCl_3} C_6H_5COCH_3$

Q.36 Answer:

$-46.32\ \text{kJ mol}^{-1}$

Solution:

For the cell reaction

$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$

$n = 2$

$F = 96500$

$E = 0.24\,V$

$\Delta G = -nFE = -2 \times 96500 \times 0.24 = -46320\,J = -46.32\,kJ$

Q.37 Answer:

1500 mg

Solution:

$1\ kg\ =\ 1000\ gm$ **.**

The total nitrogen present in $1\ kg$ **bread**

$= 1000 \times \frac{10}{100} \times \frac{7}{100}\ gm$

$\therefore$ N **converted to urea in everyday**

$= 1000 \times \frac{10}{100} \times \frac{7}{100} \times \frac{10}{100}\ gm$

$= 0.7gm$

$28\ gm\ N$ **yields** $60\ gm.$ **urea**

$\therefore\ 0.7\ gm.\ N$ yields $\frac{60}{28} \times 0.7\ gm.$ urea

$=\ 1.5\ gm\ urea.\ =\ 1500\ mg\ urea$

Q.38 Answer:

Ethyl amine

Solution:

$C_2H_5NH_2$	$\xrightarrow{HNO_2}$	C_2H_5OH	$\xrightarrow{PCl_5}$	C_2H_5Cl	$\xrightarrow{NH_3}$	$C_2H_5NH_2$
Ethyl amine		Ethyl alcohol (A)		Ethyl chloride (B)		Ethyl amine (C)

Q.39 Answer:

II

Solution:

B atom after losing outermost electron acquires noble gas configuration (stable configuration). It is difficult to remove the next electron from $B^+\left(1s^2,\ 2s^2\ 2p^6\right)$ **ion.**

Q.40 Answer:

Ethanal

Solution:

The compound which contains $-COCH_3$ group in its structure, give positive iodoform test and the compound which contains $-CHO$ group give positive Fehling test.

In ethanal, CH_3CHO both the groups are present, hence it responds to both iodoform test and Fehling's test.

Iodoform test

$$CH_3CHO + I_2 + NaOH \rightarrow CHI_3 + NaI + H_2O$$

Fehling's test

$$CH_3CHO + Cu(OH)_2 \rightarrow CH_2COOH + \underset{(\text{red ppt.})}{Cu_2O\downarrow} + 2H_2O$$

Q.41 Answer:

$T_M < T_D < T_i$

Solution:

For adiabatic expansion, $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

γ= 1.66 for mono atomic gas and 1.40 for diatomic gas.

Hence, for mono atomic gas, $\frac{T_M}{Ti} = \left(\frac{V_1}{V_2}\right)^{0.66}$

and for diatomic gas, $\frac{T_D}{Ti} = \left(\frac{V_1}{V_2}\right)^{0.40}$

As $\frac{V_1}{V_2} < 1 \left(\because V_2 > V_1\right)$**. Hence ,** $\frac{T_M}{T_i}$**<1**

i.e., $T_M < T_i$ and $\frac{T_D}{T_i} < 1$, i.e., $T_D < T_i$

suppose $V_2 = 10\ V_1$. then $\frac{T_M}{T_D} = \left(\frac{1}{10}\right)^{0.66-0.40} = \left(\frac{1}{10}\right)^{0.26}$ i.e. $T_M < T_D$ Hence $T_M < T_D < T_i$

Q.42 Answer:

Statement I is true, Statement II is also true and Statement II is the correct explanation of Statement I.

Solution:

Fluorine is the strongest oxidising agent due to its highly positive standard potential. Therefore, fluorine has the highest tendency to get reduced to F^-.

As a result, F^- ion has the least tendency to get oxidised. That is why, fluorine cannot be prepared from fluorides by chemical oxidation.

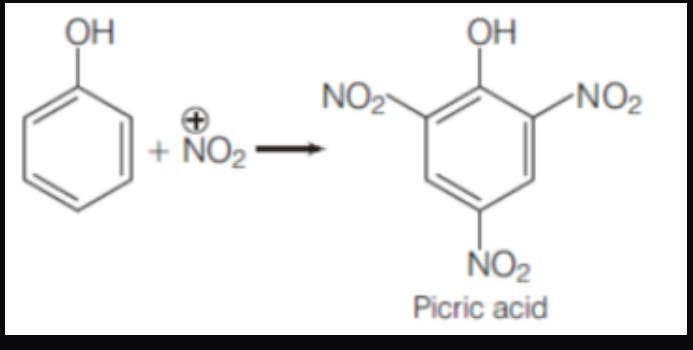
Q.43 Answer:

7 , acidic

Solution:

Phenol undergo electrophilic substitution reaction on reaction with a mixture of conc. HNO_3 and conc H_2SO_4 (known as nitrating mixture) This reaction produces nitro compound and on successive nitration it produces trinitrophenol.

conc. HNO_3 + **conc.** $H_2SO_4 \longrightarrow NO_2^+$ **(electrophile)**



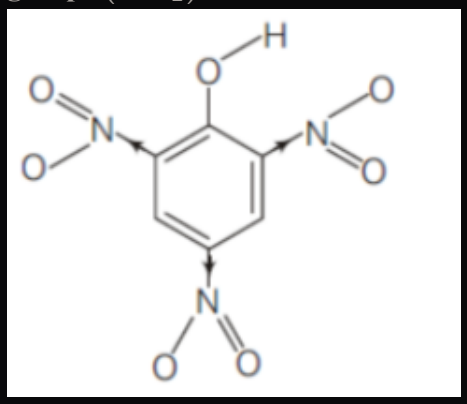
Molecular formula $= C_6H_3N_3O_7$

Degree of unsaturation

$= (C+1) - \frac{H}{2} - \frac{X}{2} + \frac{N}{2}$

$= (6+1) - \frac{3}{2} + \frac{3}{2} = 7 - 0 = 7$

Nature of compound The compound is acidic in nature due to presence of three strong electron withdrawing groups (NO_2).



Q.44 Answer:

A, *B*

Solution:

According to Huckel's Rule, a compound is said to be aromatic when it fulfills the following conditions:

The molecule is cyclic (a ring of atoms)

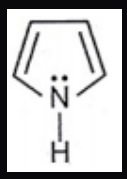
The molecule is planar (all atoms in the molecule lie in the same plane)

The molecule is fully conjugated (p orbitals at every atom in the ring)

The molecule has 4n+2 π electrons (where n can be 0 or any positive integer)

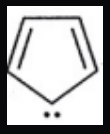
(i) Structure (a) **has two π-bonds and one lone pair of electrons over** N**- atom (**6π **electrons) and has planar structure.**

Thus, is an aromatic compound.

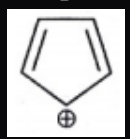


(ii) The structure (b) **also contain two π- bonds and one lone pair of electrons (**6π **electrons). Thus, follow Huckel's rule. The structure is cyclic and planar.**

$\therefore$ (b) **is also an aromatic compound.**



(iii) The structure (c) **has only** $2\pi-$ **-bonds (**$4\pi-$ **electrons) and does not follow Huckel's rule. So, it anti aromatic compound.**

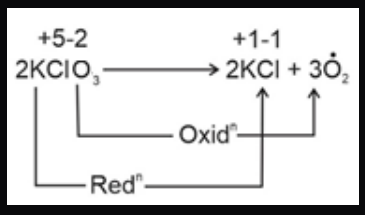


(iv) Structure (d) **has** C_{sp^3}**. So, it is non-planar hence non-aromatic.**

Q.45 Answer:

All of these

Solution:



(i, ii, iii and iv) all are false statement

Correct Option: A, B, C, D (All Options are correct)

Q.46 Answer:

—

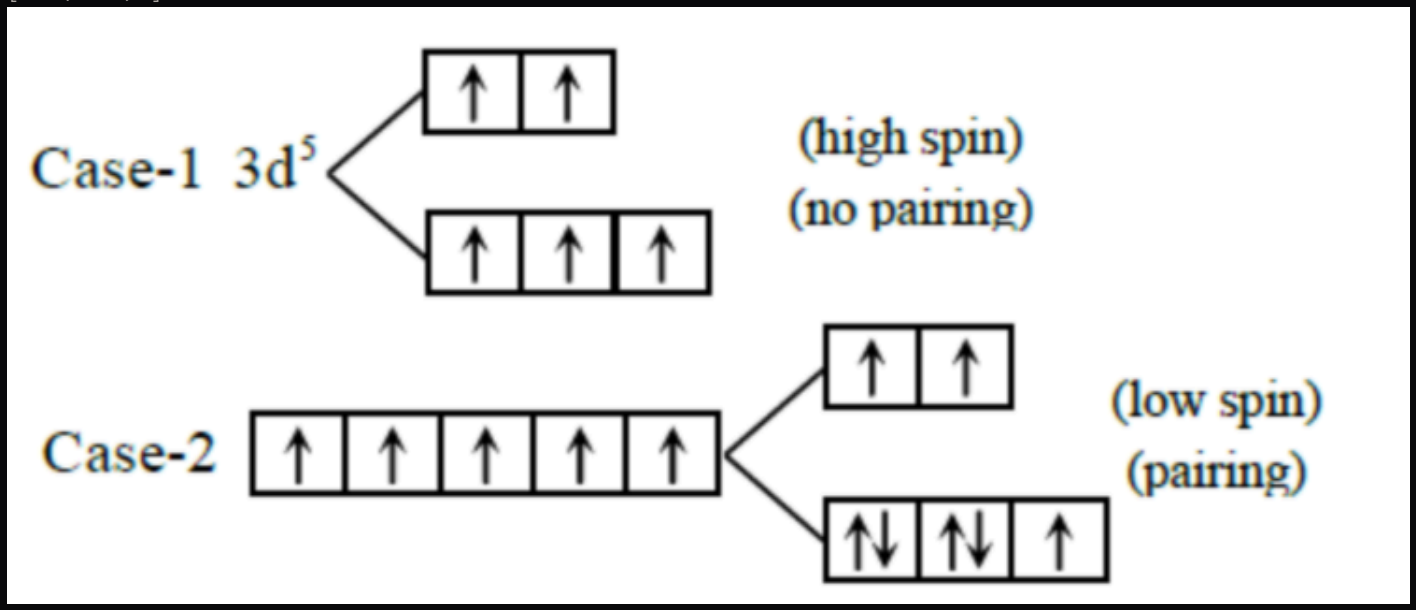
Solution:

$[\mathrm{Fe(SCN)_6}]^{3-}$ **and** $[\mathrm{Fe(CN)_6}]^{3-}$

In both the cases the electronic configuration of Fe^{3+} **will be**

$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5$

Since SCN **is a weak field ligand and** $\overline{\mathrm{CN}}$ **is a strong field ligand, the pairing will occur only in case of** $[\mathrm{Fe(CN)_6}]^3$.



Case $-1\ \mu = \sqrt{n(n+2)} = \sqrt{5(5+2)} = \sqrt{35} = 5.91\ \mathrm{BM}$

Case $-2\ \mu = \sqrt{\mathrm{n(n+2)}} = \sqrt{1(1+2)} = \sqrt{3} = 1.73\ \mathrm{BM}$

Difference in spin only magnetic moment $= 5.91 - 1.73 = 4.18 \approx 4$

Q.47 Answer:

—

Solution:

$\mathrm{K_a\ (C_6H_5COOH)} = 1 \times 10^{-4}$

$\mathrm{pH\ of\ 0.01M C_6H_5COONa}$

$\mathrm{K_h} = \frac{\mathrm{K_w}}{\mathrm{K_a}} = \frac{0.01\ \mathrm{h}^2}{1-\mathrm{h}}$

$\frac{10^{-14}}{10^{-4}} = \frac{10^{-2}h^2}{1-h}(1-h \quad 1)$
$[OH^-] = 0.01\ \text{h} = 0.01 \times 10^{-4} = 10^{-6}$
$[H^+] = 10^{-8} \text{pH} = 8$

Q.48 Answer:

—

Solution:

	$A(g) \rightarrow 2B(g) + C(s)$		
After 1st half life	$400-200$	2×200	0
After 2nd half life	$400-300$	2×300	0

So, total pressure after 48 min
$= 100 + 600 = 700\ \text{mm Hg}$

Q.49 Answer:

—

Solution:

Species Lone pair on central atom
$TeBr^{2-}$ **1**
BrF_2^+ **2**
XeF_3^- **3**

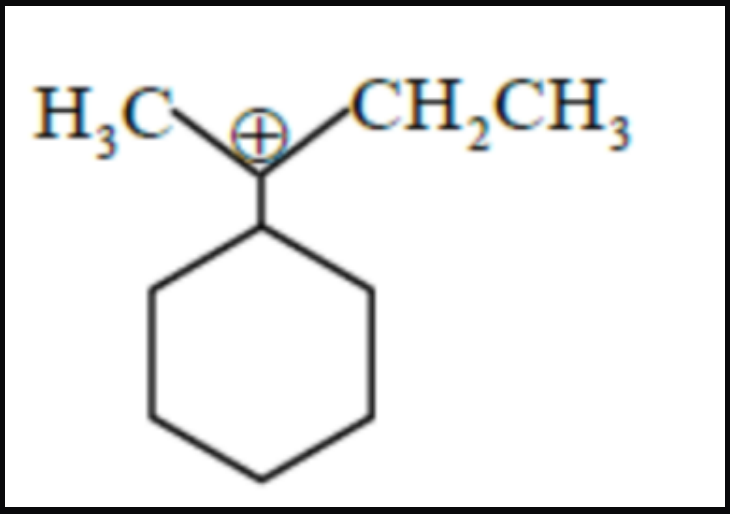
Total number of lone pair = 6.

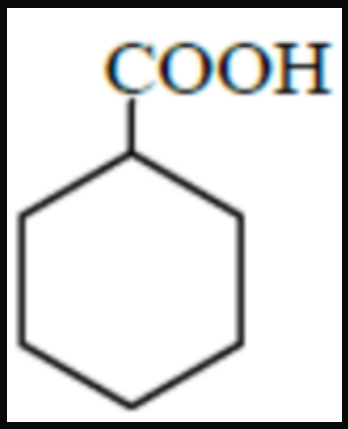
Q.50 Answer:

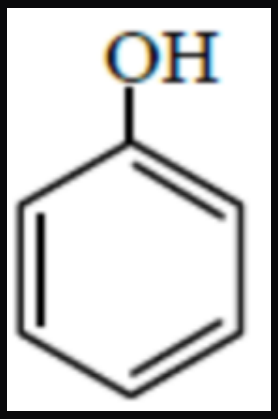
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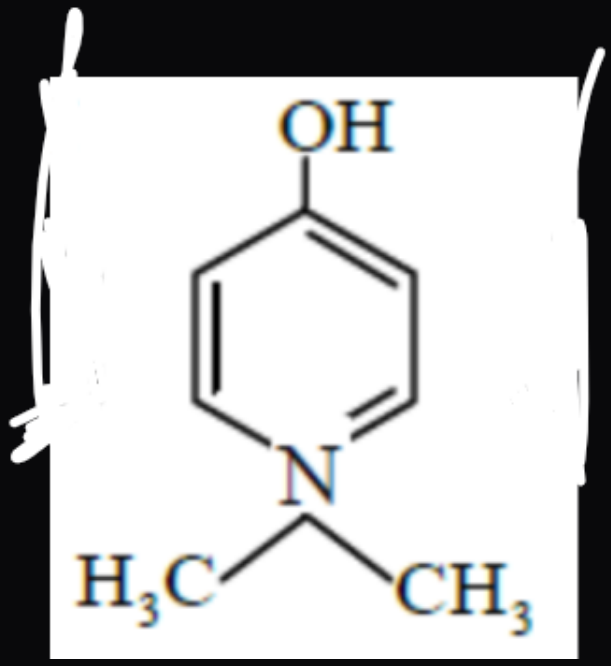
Solution:

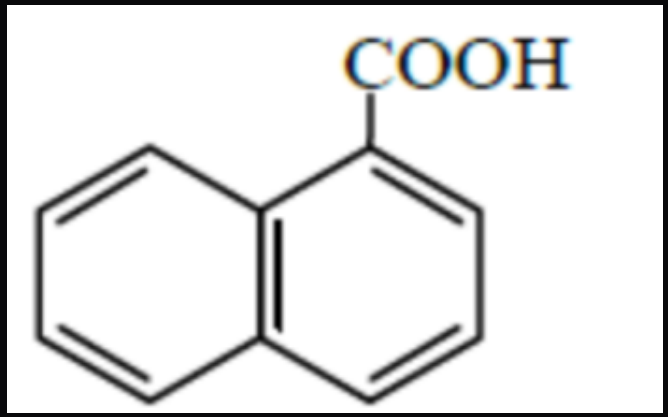
Aromatic alcohols and carboxylic acids form salt with $NaOH$**, will dissolve in aqueous** $NaOH$.











Q.51 Answer:

1

Solution:

By hypothesis, $\alpha+\beta=p;\alpha\beta=-(p+q)$

Now, $(\alpha+1)(\beta+1)=(\alpha+\beta)+\alpha\beta+1=1-q$ **.....(i)**

The given expression

$$=\frac{(\alpha+1)^2}{(\alpha+1)^2+(q-1)}+\frac{(\beta+1)^2}{(\beta+1)^2+(q-1)}$$

$$=\frac{(\alpha+1)^2}{(\alpha+1)^2-(\alpha+1)(\beta+1)}+\frac{(\beta+1)^2}{(\beta+1)^2-(\alpha+1)(\beta+1)}$$

$$=\frac{(\alpha+1)}{(\alpha-\beta)}+\frac{(\beta+1)}{(\beta-\alpha)}=\frac{(\alpha-\beta)}{(\alpha-\beta)}=1$$

Q.52 Answer:

8

Solution:

Observe that both the given lines pass through origin. So, their point of intersection will be $(0,\ 0,\ 0)$**.**

Hence, the angle bisector must pass through origin.

So, $\frac{0-2}{-1}=\frac{0+2}{1}=\frac{0+k}{4}$

$\Rightarrow k=8$

Q.53 Answer:

$(-\infty,\ 0)$

Solution:

$f'(x)=0$ **must have two real and unequal roots**

$f'(x)=3x^2-6(2\lambda-1)x+6\lambda=0$ **must have**

$D>0\Rightarrow 36(2\lambda-1)^2-72\lambda>0$

$\Rightarrow 4\lambda^2+1-4\lambda-2\lambda>0\Rightarrow 4\lambda^2-6\lambda+1>0,\ \lambda\in\left(-\infty,\ 3-\sqrt{5}\right)\cup\left(3+\sqrt{5},\ \infty\right)$

Q.54 Answer:

at most one

Solution:

Let two points $(\mathrm{x}_1,\ \mathrm{y}_1)$ **and** $(\mathrm{x}_2,\ \mathrm{y}_2)$ **lie on the circumference of a circle with centre** (π,e) **where**

$\mathrm{x}_1,\mathrm{y}_1,\mathrm{x}_2,\mathrm{y}_2\in\mathrm{Q}$

Point (x,y) **equidistant from** $(\mathrm{x}_1,\ \mathrm{y}_1)$ **&** $(\mathrm{x}_2,\ \mathrm{y}_2)$ **will lie on the perpendicular bisector of this line segment.**

$\mathrm{y}-\frac{\mathrm{y}_1+\mathrm{y}_2}{2}=-\left(\frac{(\mathrm{x}_2-\mathrm{x}_1)}{\mathrm{y}_2-\mathrm{y}_1}\right)\left(\mathrm{x}-\frac{(\mathrm{x}_1+\mathrm{x}_2)}{2}\right)$

$\Rightarrow \mathrm{px}+\mathrm{qy}+\mathrm{r}=0$ **where** $\mathrm{p},\ \mathrm{q},\ \mathrm{r}\in\mathrm{Q}$

Now it must pass through $(\pi,\mathrm{e})\Rightarrow \mathrm{p}\pi+\mathrm{qe}+\mathrm{r}=0$

$\Rightarrow \mathrm{p}=\mathrm{q}=\mathrm{r}=0$

There is no such line possible, hence atmost one point lies on the circle.

Q.55 Answer:

$9/7$

Solution:

We have,

$(3+kx)^9 = \sum_{r=0}^{9}\left\{{}^9C_r(3)^{9-r}(kx)^r\right\}$

Coefficient of x^2 = Coefficient of x^3

$\therefore {}^9C_2 3^{9-2}k^2 = {}^9C_3 3^{9-3}k^3$

$\Rightarrow 3\left(\frac{9!}{2!\cdot 7!}\right) = \left(\frac{9!}{3!\cdot 6!}\right)k$

$\Rightarrow 36 = 28k$

$\Rightarrow k = \frac{9}{7}$

Q.56 Answer:

21

Solution:

Given, number of straight lines is n.

${}^{n+5}P_{n+1} = \frac{11(n-1)}{2} \times {}^{n+3}P_n$,

$\frac{(n+5)!}{(4)!} = \frac{11(n-1)}{2} \times \frac{(n+3)!}{3!}$

$\frac{(n+5)(n+4)}{4.3.2.1} = \frac{11(n-1)}{2} \times \frac{1}{3.2}$

$(n+5)(n+4) = 11 \times 2(n-1)$

$n^2 + 9n + 20 = 22n - 22$

$n^2 - 13n + 42 = 0$

On solving above quadratic equation, we get

$n = 6$ **or** $n = 7$.

If $n = 6$, then number of points of intersection is ${}^6C_2 \times 1 = 15$.

If $n = 7$, then number of points of intersection is ${}^7C_2 \times 1 = 21$.

Now, we are getting number of points of intersection is 15 or 21. But we are having only 21 in option.

Q.57 Answer:

0.5

Solution:

$I = \int_{-1}^{2} \frac{xf(x^2)}{2+f(x+1)}dx$

$= \int_{-1}^{0} \frac{xf(x^2)}{2+f(x+1)}dx + \int_{0}^{1} \frac{xf(x^2)}{2+f(x+1)}dx$

$+ \int_{1}^{\sqrt{2}} \frac{xf(x^2)}{2+f(x+1)}dx + \int_{\sqrt{2}}^{\sqrt{3}} \frac{xf(x^2)}{2+f(x+1)}dx$

$+ \int_{\sqrt{3}}^{2} \frac{xf(x^2)}{2+f(x+1)}dx$

$= \int_{-1}^{0} 0dx + \int_{0}^{1} 0dx + \int_{1}^{\sqrt{2}} \frac{x(1)}{2+0}dx + \int_{\sqrt{2}}^{\sqrt{3}} 0dx + \int_{\sqrt{3}}^{2} 0dx$

$\Rightarrow I = \frac{1}{2}\left[\frac{x^2}{2}\right]_1^{\sqrt{2}}$

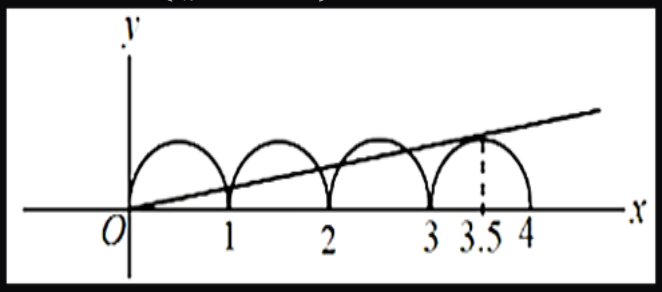
$\Rightarrow 2I = \frac{1}{2} = 0.5$

Q.58 Answer:

2

Solution:

$f(x) = \max\left\{\frac{x}{n},\ |\sin \pi x|\right\}$



Thus, for the maximum points of non-differentiability, graphs of $y = \frac{x}{n}$ and $y = |\sin \pi x|$ must intersect at maximum number of points which occurs when $n > 3.5$

Q.59 Answer:

$\frac{\pi}{3}$

Solution:

By AM-GM inequality

$\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2+x}} \geq 2 \cdot \sqrt{\sqrt{x^2 + x} \cdot \frac{\tan^2 \alpha}{\sqrt{x^2+x}}} = 2 \tan \alpha$

Since the least value of $f(x)$ is $2 \tan \alpha = 2\sqrt{3}$ (given)

Hence, $\alpha = \frac{\pi}{3}$

Q.60 Answer:

4

Solution:

The number of functions satisfying the given condition is $^9C_1 + {}^9C_2 + \ldots + {}^9C_8 = 2^9 - 2$

Probability $= \frac{2^9-2}{10^{10}} = \frac{51}{10^9}$

Q.61 Answer:

$\frac{15}{92}$

Solution:

Apart from the owner there are 24 other cars. On his returns 14 cars are left. $n(S)$ =Number of ways selecting 14 cars out of 24 cars $= {}^{24}\mathrm{C}_{14}$

$n(E) =$ **Selecting 14 cars (leaving two neighbouring positions) out of 22 cars**

$= {}^{22}C_{14}\ n\left(E\right)$ **=Selecting 14 cars (leaving two neighbouring positions) out of 22 cars**

$= {}^{22}C_{14}$

$P\left(E\right) = \frac{{}^{22}C_{14}}{{}^{24}C_{14}} = \frac{15}{92}$

Q.62 Answer:

$\mathrm{I} + \left(2^{40} - 1\right)A$

Solution:

$A^2 = \begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} = A$

$A^3 = A^4 = A^5 \ldots\ldots\ldots = A^{40} = A$

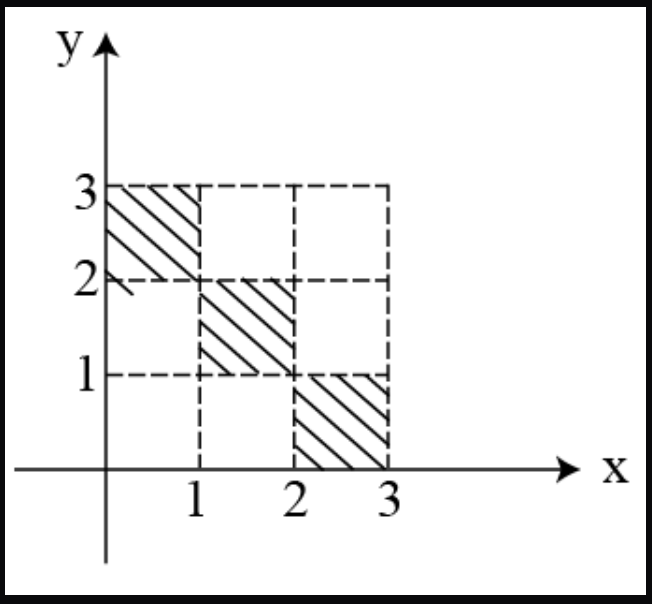
$(\mathrm{I} + A)^{40} = \mathrm{I} + {}^{40}C_1 A + {}^{40}C_2 A^2 \ldots\ldots + {}^{40}C_{40} A^{40}$

$= \mathrm{I} + A\left[{}^{40}C_1 + {}^{40}C_2 \ldots\ldots {}^{40}C_{40}\right]$
$= \mathrm{I} + A\left(2^{40} - 1\right)$

Q.63 Answer:

6

Solution:



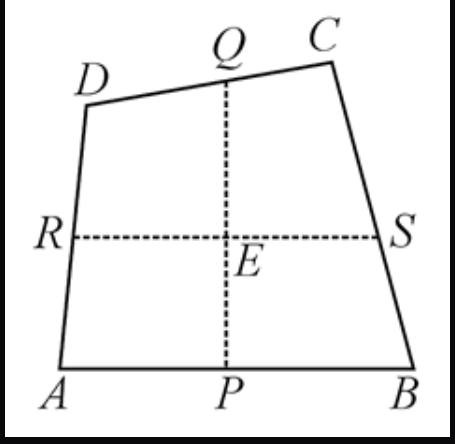
$[x] + [y] = 2$(**Let** $x, y \geq 0$)
$\Rightarrow [x], [y] = 2, 0$ **or 1,1 or 0,2**
If $[x] = 2$ **and** $[y] = 0 \Rightarrow x \in [2, 3)$ **and** $y \in [0, 1)$ **and so on Since, the curve is symmetric in the** I^{st} **and III** rd **quadrant**
Hence, the required area $= 6(1 \times 1) = 6$ **sq. units**

Q.64 Answer:

4

Solution:

Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = \vec{c}$ **and** $\overrightarrow{OD} = \vec{d}$. **Therefore,**
$\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} + \overrightarrow{OD} = \vec{a} + \vec{b} + \vec{c} + \vec{d}$
P, **the midpoint of** $\overrightarrow{AB}$, **is** $\frac{\vec{a}+\vec{b}}{2}$
Q, **the midpoint of** $\overrightarrow{CD}$, **is** $\frac{\vec{c}+\vec{d}}{2}$
Therefore, the midpoint of $\overrightarrow{PQ}$ **is** $\frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$
Similarly, the midpoint of $\overrightarrow{RS}$ **is** $\frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$, **i.e.,** $\overrightarrow{OE} = \frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4} \Rightarrow x = 4$



Q.65 Answer:

2

Solution:

$\left|(2A)^2(3B)(5C)^{-1}\right| = \frac{|2A|^2|3B|}{|5C|} = \frac{2^{2n}3^n}{5^n}\frac{|A|^2|B|}{|C|}$
$\Rightarrow \left(\frac{12}{5}\right)^{n+1} = \left(\frac{12}{5}\right)^3 \Rightarrow n = 2$

Q.66 Answer:

$(1,\ 3)$

Solution:

Let A $(-2,\ -6)$, B$(-2,\ 4)$ **and** C$(1,\ 3)$
$\mathrm{AB} = \sqrt{(-2+2)^2 + (4+6)^2} = 10$
$\mathrm{BC} = \sqrt{(1+2)^2 + (3-4)^2} = \sqrt{10}$
$\mathrm{CA} = \sqrt{(-2-1)^2 + (-6-3)^2} = \sqrt{90}$
$(\mathrm{AB})^2 = (\mathrm{BC})^2 + (\mathrm{CA})^2$
Right angle at C
$\Rightarrow$ orthocentre $(1,\ 3)$

Q.67 Answer:

None of these

Solution:

$\frac{1}{1+i} + \frac{1}{1-i} + \frac{1}{-1+i} + \frac{1}{-1-i} + \frac{2}{1+i} + \frac{2}{1-i} + \frac{2}{-1+i} + \frac{2}{-1-i} + \frac{3}{1+i} + \frac{3}{1-i} + \frac{3}{-1+i} + \frac{3}{-1-i} + \ldots + \frac{n}{1+i} + \frac{n}{1-i} + \frac{n}{-1+i} + \frac{n}{-1-i}$
$= \left(\frac{1}{1+i} - \frac{1}{i-1} + \frac{1}{i-1} - \frac{1}{i+1}\right) + \left(\frac{2}{i+1} - \frac{2}{i-1} + \frac{2}{i-1} - \frac{2}{i+1}\right) \cdots\cdots + \left(\frac{n}{1+i} - \frac{n}{i-1} + \frac{n}{i-1} - \frac{n}{i+1}\right)$
On cancelling the terms, we get
$= 0$

Q.68 Answer:

-1

Solution:

Let, $L = \lim_{x \to \frac{\pi}{4}} \frac{\tan x - 1}{\cos 2x}$
$= \lim_{h \to 0} \frac{\tan\left(\frac{\pi}{4}+h\right) - 1}{\cos 2\left(\frac{\pi}{4}+h\right)}$
(Putting $x = \frac{\pi}{4} + h$**)**

$= \lim_{h\to 0} \frac{\left(\frac{1+\tan h}{1-\tan h}\right)-1}{\cos\left(\frac{\pi}{2}+2h\right)}$

$= \lim_{h\to 0} \frac{1+\tan h-1+\tan h}{-\sin 2h(1-\tan h)}$

$= \lim_{h\to 0} \frac{-2\tan h}{2\sin h\cos h\,(1-\tan h)}$

$= \lim_{h\to 0} \frac{-1}{\cos^2 h(1-\tan h)}$

$= -1.$

Q.69 Answer:

$\frac{2}{9}$

Solution:

$\ell_1\ell_2 + \mathrm{m}_1\mathrm{m}_2 + \mathrm{n}_1\mathrm{n}_2 = 0$

$\Rightarrow \frac{1}{2} + \frac{1}{\lambda} - 5 = 0$

Q.70 Answer:

$-YX$

Solution:

Assume A and B are symmetric matrices & C is skew-symmetric matrix.

$A^T = A,\ B^T = B,\ C^T = -C$

Now,

$X = ABC + CBA$

$Y = ABC - CBA$

Let

$D = (XY)^T = Y^T X^T$

$\Rightarrow D = (ABC - CBA)^T (ABC + CBA)^T$

$\Rightarrow D = \left[(ABC)^T - (CBA)^T\right]\left[(ABC)^T + (CBA)^T\right]$

$\Rightarrow D = \left[C^T B^T A^T - A^T B^T C^T\right]\left[C^T B^T A^T + A^T B^T C^T\right]$

$\Rightarrow D = [-C\,B\,A + A\,B\,C][-C\,B\,A - A\,B\,C]$

$\Rightarrow D = -YX.$

Q.71 Answer:

—

Solution:

$\mathrm{I} = \int_0^{\pi} \sqrt{\begin{matrix}\cos^2 x + \cos^2(2x) + \cos^2 3x + 2(\cos x\cos 2x + \cos 2x\cos 3x + \cos x\cos 3x) \\ +\sin^2 x + \sin^2 2x + \sin^2 3x + 2(\sin x\sin 2x + \sin 2x\sin 3x + \sin x\sin 3x)\end{matrix}}$

$= \int_0^{\pi} \sqrt{3 + 2(2\cos\mathrm{x} + \cos 2\mathrm{x})}\mathrm{dx}$

$= \int_0^{\pi} \sqrt{3 + 4\cos\mathrm{x} + 2\cos 2\mathrm{x}}\ \mathrm{dx}$

$= \int_0^{\pi} \sqrt{1 + 2(1 + \cos 2\mathrm{x}) + 4\cos\mathrm{x}}\ \mathrm{dx}$

$= \int_0^{\pi} \sqrt{1 + 2.2\cos^2 \mathrm{x} + 4\cos\mathrm{x}}\ \mathrm{dx}$

$= \int_0^{\pi} \sqrt{4\cos^2 \mathrm{x} + 4\cos\mathrm{x} + 1}\ \mathrm{dx}$

$= \int_0^{\pi} |2\cos\mathrm{x} + 1|\mathrm{dx} = 2\sqrt{3} + \frac{\pi}{3}$

$= \frac{\pi}{3} + \sqrt{12}$

Compare with $\left(\frac{\pi}{\mathrm{k}} + \sqrt{\mathrm{w}}\right)$.

Thus,

$\mathrm{k}=3;\ \mathrm{w}=12$

So,

$\mathrm{k}^2+\mathrm{w}^2=153$

Q.72 Answer:

—

Solution:

The number of real values of m for .....

$A=\{x:(x-2)(x+m+1)=0,\ x\in R\}$

$B=\ \{x:(m-1)x^2+mx+1=0,\ x\in R\}$

Case I.

When $m-1=0 \qquad\qquad m=1$

$A=\{2,\ -2\}$ **and** $B=\{-1\}$

$\therefore\ A\cup B=\{2,\ -2,\ -1\}\Longrightarrow$ **exactly 3 elements**

$\therefore\ m=1$

Case - II

When $m\neq 1$

$A=\{2,\ -m-1\},\ B=\left\{-1,\frac{-1}{m-1}\right\}$

$2=-m-1\ \Rightarrow\ \ m=-3$

$A=\{2\},\ B\{-1,\frac{1}{4}\}$

$\therefore\ m=-3$

$2=\frac{-1}{m-1}\ \Rightarrow\ \ m=\frac{1}{2}$

$A=\left\{2,\frac{3}{2}\right\},\ B=\{-1,2\}$

$\therefore\ m=\frac{1}{2}$

$-m-1=\frac{-1}{m-1}\ \ \Rightarrow\ \ m^2-1=1$

$\therefore\ m=\pm\sqrt{2}$

$-m-1=-1\ \ \Rightarrow\ \ m=0$

$-1=\frac{-1}{m-1}\ \Rightarrow\ m-1=1$

$\therefore\ m=2$

Number of values of $m=7$

Q.73 Answer:

—

Solution:

Circle having foci of the hyperbola S_1 **as the extremities of diameter should not intersect the hyperbola** S_2 **at real points**

$\Rightarrow a^2+b^2<4a^2$

$\Rightarrow b^2<3a^2$

$\Rightarrow e^2<7/4$

$\Rightarrow 4<4e^2<7$

Q.74 Answer:

—

Solution:

Given,

$\mathrm{F(mn)} = \mathrm{f(m)} \cdot \mathrm{f(n)}$

Put $\mathrm{m} = 1$**, then** $\mathrm{f}(1 \cdot \mathrm{n}) = \mathrm{f}(1) \cdot \mathrm{f(n)} \Rightarrow \mathrm{f}(1) = 1 \quad \{f(n) \neq 0\}$

Put $\mathrm{m} = \mathrm{n} = 2$

$\mathrm{f}(4) = \mathrm{f}(2) \cdot \mathrm{f}(2)$

If $f(2) = 1$**, then** $f(4) = 1$ **and if** $f(2) = 2$**, then** $f(4) = 4$

Put $\mathrm{m} = 2,\ \mathrm{n} = 3$

$f(6) = f(2) \cdot f(3)$

If $f(2) = 1$**, then** $f(3)$ **can be any value between** 1 **to** 7

If $f(2) = 2$**, then** $f(3)$ **can be** $1\ \text{or}\ 2\ \text{or}\ 3$

$\mathrm{f}(5),\ \mathrm{f}(7)$ **can take any value from** $\{1, 2, \ldots, 7\}$

So, total number of function $= (1 \times 1 \times 7 \times 1 \times 7 \times 1 \times 7) + (1 \times 1 \times 3 \times 1 \times 7 \times 1 \times 7)$

$= 490$

Q.75 Answer:

—

Solution:

$T_1 = 2014$

$T_2 = (2)^3 + (0)^3 + (1)^3 + (4)^3 = 73$

$T_3 = (7)^3 + (3)^3 = 370$

$T_4 = (3)^3 + (7)^3 + (0)^3 = 370$

$T_5 = (3)^3 + (7)^3 + (0)^3 = 370$

$T_{2014} = (3)^3 + (7)^3 + (0)^3 = 370$